

To amend the Federal Food, Drug, and Cosmetic Act to require an automated verification, validation, and uncertainty quantification process to clear robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation.

IN THE HOUSE OF REPRESENTATIVES

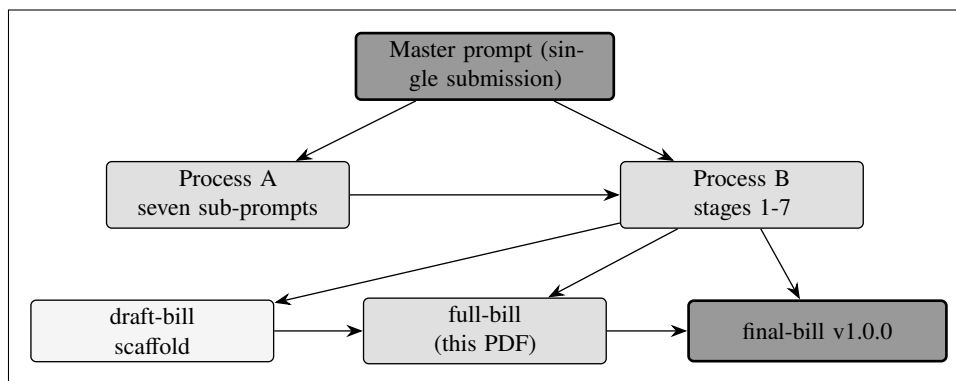
June 7, 2026

Mr. Kevin Kawchak authored the following bill with the assistance of artificial intelligence; which was referred to the Committee on Energy and Commerce.

A BILL

This legislative draft is submitted for review and consideration by the United States Congress,

Bill (v4.0) 10.5281/zenodo.xxxxxxxx is the polished, final gray-scale Mermaid amendment of H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026. It renders nine gray-scale Mermaid figures and eleven full-width tables, carries the full operative amendment, and writes all twelve submission deliverables as LaTeX appendices, each in the clickable table of contents. It was prepared with an autonomous artificial intelligence coding agent (Claude Code Opus 4.8, 1M context, Max) under human direction. It is an independent research draft, is not an enacted law, and is not legal advice; the development record appears in the provenance back matter.



Cover Figure. The Bill v4.0 single-prompt build: one Master prompt generates every sub-prompt (Process A), which run in sequence (Process B) to grow the bill from the draft scaffold to this full bill and the polished final bill.

SECTION 1. SHORT TITLE; TABLE OF CONTENTS.

(a) **Short Title.** This Act may be cited as the “Verification Before Generation in Physical AI Oncology Trials Act of 2026”.

(b) **Table of Contents.** The table of contents for this Act version is as follows. Each entry links to the part it names.

Sec. 1. Short title; table of contents.	1
<i>The citation and this clickable contents.</i>	
Sec. 2. Findings; the evidence to law record.	2
<i>Why the rule is needed and feasible, with the four-work record.</i>	
Sec. 3. Amendment to the Federal Food, Drug, and Cosmetic Act.	5
<i>New section 515D and ten conforming amendments.</i>	
Sec. 4. Comparative print; changes in existing law.	9
<i>The eleven affected Title 21 device sections, marked.</i>	
App. A. One-page summary.	11
<i>The problem, the rule, the evidence, and the ask, on one page.</i>	
App. B. Section-by-section analysis.	12
<i>Plain-English gloss of every operative part.</i>	
App. C. Plain-English policy memo.	15
<i>Why the sequencing gap matters and why the gate fits existing law.</i>	
App. D. Legislative findings.	17
<i>The fourteen findings of fact, each tied to an in-force authority.</i>	
App. E. Ramseyer comparative print.	19
<i>Changes to existing law, marked, for the affected sections.</i>	
App. F. Constitutional Authority Statement.	21
<i>The Commerce Clause basis (House Rule XII).</i>	
App. G. PAYGO and cost estimate.	22
<i>PAYGO note and a negative earmark declaration.</i>	
App. H. Sponsor and cosponsor packet.	24
<i>The sponsor role and the cosponsor form.</i>	
App. I. Stakeholder engagement plan.	26
<i>Outreach to agencies, advocacy, industry, and standards bodies.</i>	
App. J. Legislative Counsel routing memo.	28
<i>The legislative-form questions for HOLC.</i>	
App. K. Currency and cross-reference matrix.	32
<i>Re-verification that every citation still lands.</i>	
App. L. Testimony and research-influence brief.	35
<i>Hearing testimony; emerging bills kept out of the text.</i>	

SEC. 2. FINDINGS; THE EVIDENCE TO LAW RECORD.

(a) **In General.** Congress finds that an automated process must verify robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation, and that the record of the VVUQ-01 through VVUQ-04 developments, visualized in this section, establishes both the need for and the feasibility of that requirement [1, 2]. The lineage in Figure 1 shows how engineering evidence became proposed law and then a precise amendment.

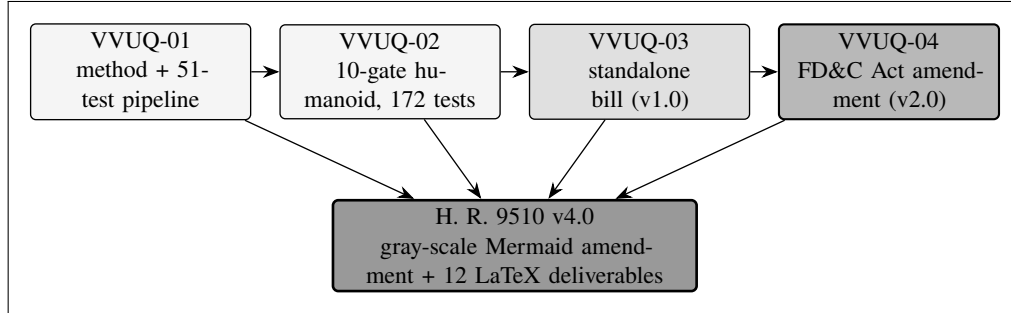


Figure 1. Four-work evidence-to-law lineage. The method and pipeline (VVUQ-01), the hard engineering proof (VVUQ-02), the proposed bill (VVUQ-03), and the precise amendment (VVUQ-04), feeding this v4.0 amendment.

(b) **The Embodiment Problem and the Sequencing Gap.** Physical AI oncology investigations place autonomous and semi-autonomous robots in direct physical contact with patients, and the behavior of a robot at the moment of contact is determined by software that is increasingly machine-generated. In a disembodied pipeline an error yields a wrong number a human can catch downstream; in an embodied surgical system the same error class yields a wrong motion a human cannot catch in time. No Federal law requires the verification of robot-patient interaction code to clear before that code is generated or executed, because the order of operations is not regulated [3, 1]. Table 1 records what each of the four works produced.

Work	What it is	Releases	Headline assurance result	End-goal DOI
VVUQ-01	Method paper plus working pipeline and an execution record, holding VVUQ higher than code generation	v0.1.0 to v0.6.0	51 of 51 tests; a 1 ACCEPT, 5 BLOCK, 1 ESCALATE gate surface; 2.5x acceleration	zenodo.xxxxxxxx
VVUQ-02	Generated ten-gate humanoid surgical codebase, autonomous execution record, and manuscript	v0.7.0 to v1.2.0	172 of 172 tests; 10 gates; 32 of 32 sweep iterations clear; composite mean 93.56	zenodo.xxxxxxxx
VVUQ-03	Standalone Physical AI Oncology Trial Bill (H. R. 9510), synthesized from the recorded evidence	v1.3.0 to v1.4.0	15 sections; 21 body-width tables; 60 references; verification before generation as statute	zenodo.xxxxxxxx
VVUQ-04	Bill recast as an amendment to the Federal Food, Drug, and Cosmetic Act, through Public Law 119-93	v2.0.0 to v2.3.0	New section 360e-5 plus ten conforming amendments; legal crosswalk current to May 31, 2026	zenodo.xxxxxxxx

Table 1. The four works at a glance, with each work's role, release range, headline assurance result, and DOI.

- (c) **Feasibility Is Demonstrated.** Automated verification of robot-patient interaction code is feasible today. A pipeline held the verification fraction at 1.0, accepting one candidate artifact and blocking five, and a ten-gate assurance suite over control software for a surgical humanoid cleared reproducibly from a fixed seed, each gate bound to a published external standard [3, 4].
- (d) **The Closest Existing Analogues.** The predetermined change control plan authority in section 515C of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 360e-4) and the Food and Drug Administration final guidance on such plans for artificial-intelligence-enabled device software functions are the closest existing analogues to validating a change before deployment, but each remains voluntary, submission-based, and neither automated nor required in advance of generation. The Quality Management System Regulation (part 820 of title 21, Code of Federal Regulations), effective February 2, 2026, and the electronic-records requirements of part 11 of such title supply quality and record-integrity scaffolding but with no rule that verification precedes generation [5, 6].
- (e) **A Recognized Standards Basis.** Recognized consensus standards, including ASME V&V 40-2018, the Food and Drug Administration guidance assessing the credibility of computational modeling, NASA-STD-7009A, IEEE Std 7009-2024, IEC 80601-2-77, ISO 14971, and ISO/TS 15066, supply the credibility framework, dispersion bounds, fail-safe requirements, and biomechanical contact limits to which an automated gate may be bound [7, 8, 9, 10].
- (f) **Human-Over-AI Review Is Already Widely Adopted.** Human-over-AI review is already widely adopted in State law, including the California Physicians Make Decisions Act (Senate Bill 1120), the Illinois Wellness and Oversight for Psychological Resources Act (House Bill 1806), the Texas Responsible Artificial Intelligence Governance Act (House Bill 149), and Maryland House Bill 820, and the strongest Federal precedent for mandatory human-over-AI oversight is the Medicare Advantage artificial-intelligence guardrail (section 422.101(c) of title 42, Code of Federal Regulations) [11, 12, 13, 14, 15].
- (g) **Modernization, Nondiscrimination, and Classification.** As clinical investigation moves toward real-time and decentralized designs governed by harmonized good clinical practice, manual review cannot keep pace with the volume of machine-generated control code, and an automated gate that clears before generation serves that modernization. The acceleration the four works show is set against the conventional path in Figure 2. Federal nondiscrimination law, including section 92.210 of title 45, Code of Federal Regulations, title VI of the Civil Rights Act of 1964 (42 U.S.C. 2000d), and the Americans with Disabilities Act of 1990 (42 U.S.C. 12101 et seq.), requires that a patient-facing algorithm minimize the risk of bias and respect accessibility, and the artificial-intelligence taxonomy maintained by the American Medical Association provides a graduated spine for matching verification rigor to the degree of autonomy of a device [16, 17, 18, 19, 20].

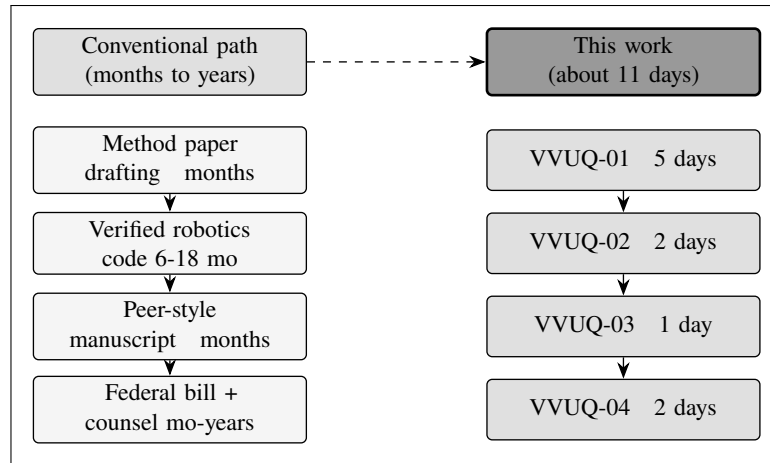


Figure 2. Accelerated timeline versus conventional methods. The conventional months-to-years path beside the about eleven days of autonomous, real-time work that produced VVUQ-01 through VVUQ-04.

SEC. 3. AMENDMENT TO THE FEDERAL FOOD, DRUG, AND COSMETIC ACT.

(a) In General. Subchapter V of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 351 et seq.) is amended by inserting after section 515C (21 U.S.C. 360e-4) the following new section:

“SEC. 515D. VERIFICATION BEFORE GENERATION OF ROBOT-PATIENT INTERACTION CODE IN PHYSICAL AI ONCOLOGY INVESTIGATIONS.

“(a) Requirement.- No robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated verification, validation, and uncertainty quantification process, bound to one or more named external standards, has first cleared for that interaction, and the cleared verification record has been documented and attested under subsection (e).

“(b) Order of Operations.- A sponsor shall not generate, and shall not execute, robot-patient interaction code in advance of the cleared verification record. A record created after generation or execution does not satisfy this section.

“(c) Gate Thresholds and Decision Rule.- For each robot-patient interaction, the verification process shall apply the gate thresholds set out in the schedule in Table 2. Each gate shall emit a determination of ACCEPT, BLOCK, or ESCALATE; a single failing dimension shall set BLOCK, and ambiguity or divergence shall set ESCALATE and return the interaction to a qualified human. For each interaction the verification fraction shall equal 1.0, and, for an interaction subject to a hard catastrophe predicate, that predicate shall also hold [7, 9].

Gate	Min. agreement	Max. CV	Hard predicate	Governing external standards
01 Bimanual hand-eye servo	0.97	0.08	No	IEC 80601-2-77; ISO 9283; ASME V&V 40
02 Dexterous finger force	0.95	0.10	No	IEC 80601-2-77; ISO/TS 15066
03 Whole-body balance	0.98	0.06	No	ISO 13482; IEC 60601-1
04 Auton. plan correct	0.95	0.10	No	UL 4600; IEEE 7009; IEC 62304
05 Instr. grasp, handover	0.96	0.10	No	IEC 80601-2-77; ISO 9283
06 Vasc. no-fly vol. (hand)	1.00	0.05	Yes	IEC 80601-2-77; ISO 14971
07 Bimanual suturing anast.	0.96	0.08	No	IEC 80601-2-77; Fistula Risk Score
08 Perception scene unders.	0.95	0.10	No	ASME V&V 40; IEC 62304
09 Shared-room human coll.	1.00	0.06	Yes	ISO/TS 15066; ISO 10218-1; ISO 13482
10 Fault, emer. stop, degrad.	1.00	0.05	Yes	IEC 60601-1; ISO 13849-1; IEEE 7009

Table 2. Ten-gate threshold schedule: min. validation agreement, max. coefficient of variation, hard catastrophe predicate, and external standards for each gate. Gates 06, 09, and 10 are hard predicate at agreement 1.00.

The decision rule and the gate funnel that this schedule implements are shown in Figure 3.

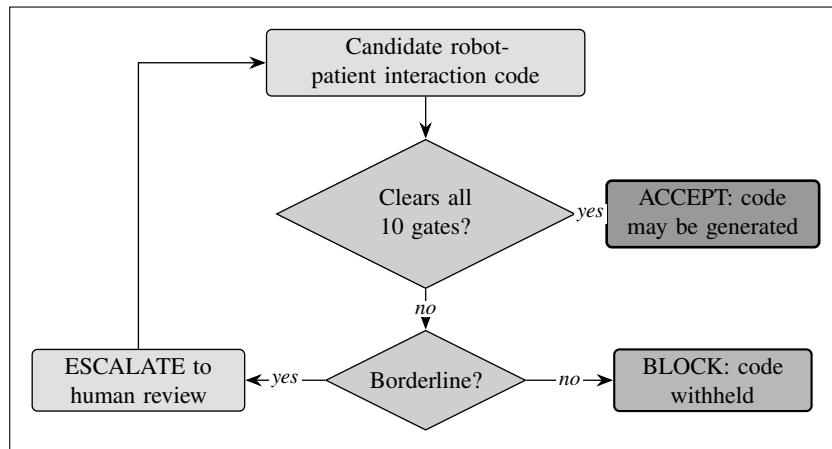


Figure 3. The VVUQ gate decision rule over the verify, validate, and quantify dimensions: a single failing dimension sets BLOCK, divergence sets ESCALATE to a qualified human, and only a full clear sets ACCEPT.

“(d) Readiness Gates.- Before a patient is enrolled or a robot acts, the trial site shall pass the Physical AI Standard Level gate on legislative authorization, regulatory compliance, and operational readiness; each robot shall meet the minimum Unification Standard Level for its functional type, which is not less than 7.0 for a surgical type, 6.0 for a therapeutic positioning or diagnostic needle placement type, 4.0 for a rehabilitative type, and 3.0 for a companion monitoring type; and each robot’s behavior shall be validated across not fewer than two simulation frameworks within the cross-framework tolerances of less than 2 millimeters of trajectory deviation and less than 0.5 newton of force discrepancy [1].

“(e) Documentation and Attestation.- Before robot-patient interaction code is generated or executed, the sponsor shall make and file, for each covered algorithm, algorithm documentation including a plain-language summary, the decision logic or architecture, the training or conditioning data sets, the gate bindings identifying the governing external standard and the threshold enforced, the verification-before-generation record, and a determinism statement giving the seed; a compliance statement and a signed attestation by a responsible officer; and a hash-chained, tamper-evident audit trail that satisfies part 11 of title 21, Code of Federal Regulations [21, 22].

“(f) Cybersecurity, Human Oversight, and Lifecycle.- Each Physical AI system shall incorporate authentication and access control, encryption of data in transit and at rest, network segmentation, and software-integrity verification; shall operate under recorded human oversight with a manual override and an emergency stop that brings the system to a safe state within the procedure-appropriate budget and a hand-back-to-human escalation when a gate escalates or a fault is detected; and shall maintain configuration and model-version management, monitor for model and data drift, and, on decommissioning, archive the audit trail, securely erase patient data, and revoke access [23, 24].

“(g) Nondiscrimination.- Each covered algorithm and its training data shall minimize the risk of bias on protected characteristics and adhere to evidence-based clinical guidelines, and each Physical AI system shall be designed and operated consistent with applicable accessibility and accommodation requirements [17, 18, 19].

“(h) Regulations.- Not later than 365 days after the date of enactment of this section, the Secretary shall issue final regulations to carry out this section. Pending those regulations, the Secretary may issue interim guidance, and a sponsor may comply by meeting the requirements of subsections (a) through (g).

“(i) Definitions.- In this section, the terms ‘Physical AI system’, ‘robot agent’, ‘robot-patient interaction code’, ‘generative artificial intelligence’, ‘AI-enabled device software function’, ‘verification’, ‘validation’, ‘uncertainty quantification’, ‘verification fraction’, ‘validation agreement’, ‘coefficient-of-variation bound’, ‘hard catastrophe predicate’, ‘Physical AI Standard Level’, ‘Unification Standard Level’, and ‘hash-chained audit trail’ have the meanings given in the prior bill and reconciled with section 201(h) and section 520(o) of this current Act.

“(j) Rule of Construction.- Nothing in this section displaces the investigational-device exemption under section 520(g), the protection of human subjects under part 50 of title 21, Code of Federal Regulations, the investigational new drug framework under part 312 of such title, or the predetermined change control plan authority under section 515C as it applies outside Physical AI, and nothing in this section itself reclassifies any device.”.

(b) Conforming Amendments. The Federal Food, Drug, and Cosmetic Act is amended, with the corresponding changes shown in the comparative print in section 4, so that the device definition in section 201(h) reaches software that generates or executes robot-patient interaction code; the prohibited acts in section 301 reach a violation of section 515D; a device used without a cleared section 515D record is adulterated under section 501; the real world evidence program in section 505F recognizes section 515D records; a verification-governed change consistent with a predetermined change control plan needs no new premarket notification under section 510(k); special controls under section 513 are graduated by degree of autonomy; a premarket approval application under section 515 must include the cleared verification record; a predetermined change control plan under

section 515C must include an automated verification-before-change protocol; the clinical-decision-support exclusion in section 520(o) does not reach robot-control software; and a savings clause at section 521 preserves State human-in-the-loop requirements that are not different from, or in addition to, a Federal device requirement [25, 26, 27, 28, 29, 30, 31, 32, 33, 34]. Figure 4 shows where the new section attaches on the device-regulation pathway, and Table 3 is the conforming-amendment crosswalk.

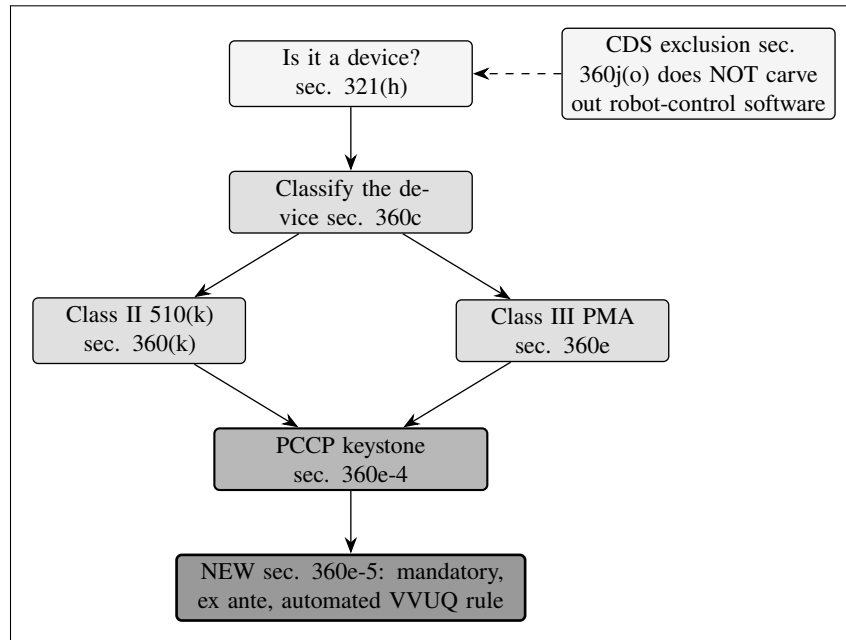


Figure 4. Statutory layering of new section 360e-5 through Title 21: the device question, the clinical-decision-support carve-out it must not fall through, classification, the 510(k) and premarket-approval pathways, the section 360e-4 keystone, and the new mandatory, ex ante section 360e-5.

No.	FD&C §	21 U.S.C.	Conforming change made by H. R. 9510
(1)	§ 201(h)	§ 321(h)	Device definition reaches software that generates or executes robot-patient code
(2)	§ 301	§ 331	New prohibited act for generation or execution in violation of section 515D
(3)	§ 501	§ 351	Device used without a cleared section 515D record is adulterated
(4)	§ 505F	§ 355g	Real world evidence program recognizes section 515D records
(5)	§ 510(k)	§ 360(k)	Verification-governed change needs no new premarket notification
(6)	§ 513	§ 360c	Special controls graduated by degree of autonomy
(7)	§ 515	§ 360e	Premarket approval application must include the cleared verification record
(8)	§ 515C	§ 360e-4	Change control plan must include automated verification-before-change protocol
(9)	§ 520(o)	§ 360j(o)	Clinical-decision-support exclusion does not reach robot-control software
(10)	§ 521	§ 360k	Savings clause preserves State human-in-the-loop requirements

Table 3. The ten conforming amendments crosswalk: each amendment by its Federal Food, Drug, and Cosmetic Act section, its 21 U.S.C. citation, and the change H. R. 9510 makes. The text appears in comparative print section 4.

- (c) **Clerical Amendment.** The table of contents for the Federal Food, Drug, and Cosmetic Act is amended by inserting after the item relating to section 515C the following new item:
- “Sec. 515D. Verification before generation of robot-patient interaction code in Physical AI oncology investigations.”.
- (d) **Rule of Construction.** Nothing in this Act displaces the investigational-device exemption under section 520(g) of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 360j(g)), the protection of human subjects under part 50 of title 21, Code of Federal Regulations, the investigational new drug framework under part 312 of such title, the predetermined change control plan authority under section 515C as it applies outside Physical AI, or any State law preserved under section 521 as amended; and nothing in this Act reclassifies any device [35, 36].
- (e) **Effective Date; Implementation Deadline; Transition Rule.** This Act takes effect on the first day of the first calendar quarter beginning not less than one year after enactment; the Secretary shall issue the final regulations required by section 515D(h) not later than 365 days after enactment; and a Physical AI oncology clinical investigation already enrolling on the effective date shall have 180 days to come into compliance with section 515D, and this Act does not disturb the February 2, 2026, effective date of the Quality Management System Regulation [6].

SEC. 4. COMPARATIVE PRINT; CHANGES IN EXISTING LAW.

(a) **Reader’s Note on Conventions.** This section shows, in comparative print, the changes the amendment makes to existing law. Existing text is plain; matter proposed to be struck is shown [strike: a strike marker]; matter proposed to be inserted is shown [insert: *an insert marker*]. No strikethrough or underline is used. Table 4 is the per-section change map, and Figure 5 shows the eleven affected sections in comparative-print order.

FD&C §	21 U.S.C.	Nature of the marked change
new § 515D	§ 360e-5	Insert the entire new section (verify before generation)
§ 201(h)	§ 321(h)	Insert reach to robot-patient interaction code
§ 301	§ 331	Insert a new prohibited act paragraph
§ 501	§ 351	Insert an adulteration trigger for a missing 515D record
§ 505F	§ 355g	Insert recognition of 515D records in the RWE program
§ 510(k)	§ 360(k)	Insert a verification-governed-change exception
§ 513	§ 360c	Insert graduated special controls by degree of autonomy
§ 515	§ 360e	Insert the cleared record into the PMA contents
§ 515C	§ 360e-4	Insert an automated verification-before-change protocol
§ 520(o)	§ 360j(o)	Insert a carve-back excluding robot-control software
§ 521	§ 360k	Insert a State human-in-the-loop savings clause

Table 4. Per-section comparative-print change map: the new section and the ten conforming amendments, each by Federal Food, Drug, and Cosmetic Act section, 21 U.S.C. citation, and the nature of the marked change.

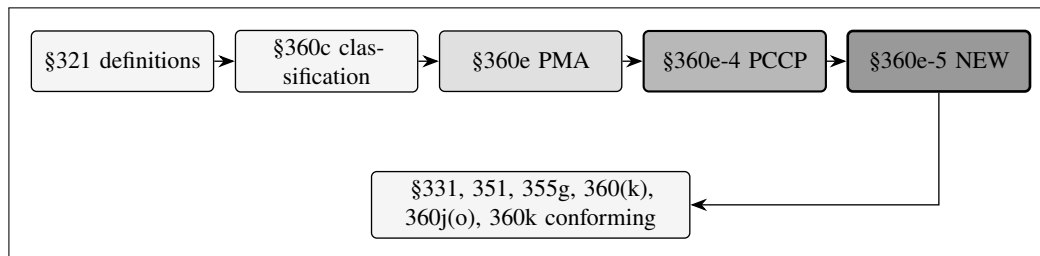


Figure 5. The eleven affected sections in comparative-print order: the classification-to-approval spine, the new section 360e-5, and the conforming amendments threaded through the rest of Title 21.

(b) **New Section 515D (21 U.S.C. 360e-5).** The following new section is inserted after section 515C:

[insert: *Sec. 515D. Verification before generation of robot-patient interaction code in Physical AI oncology investigations. (a) No robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated verification, validation, and uncertainty quantification process, bound to one or more named external standards, has first cleared for that interaction, and the cleared verification record has been documented and attested. The full text of subsections (a) through (j) is set out in section 3 of this Act.*]

(c) **Section 201(h) (21 U.S.C. 321(h)); Device.**

“The term ‘device’ means an instrument, apparatus, implement, machine, ... including any component, part, or accessory, [insert: *including software that generates or executes robot-patient interaction code in a Physical AI oncology clinical investigation,*] which is ... intended to affect the structure or any function of the body ...”.

(d) **Section 301 (21 U.S.C. 331); Prohibited Acts.**

“The following acts and the causing thereof are prohibited: [insert: *(fff) The generation or execution of robot-patient interaction code in a Physical AI oncology clinical investigation in violation of section 515D.*]”.

(e) Section 501 (21 U.S.C. 351); Adulterated Devices.

“A drug or device shall be deemed to be adulterated ... [insert: *(k) if it is a device used in a Physical AI oncology clinical investigation for which a cleared verification record required by section 515D has not been documented and attested before the robot-patient interaction code was generated or executed.*].”.

(f) Section 510(k) (21 U.S.C. 360(k)); Premarket Notification.

“... a report under this subsection [insert: *is not required for a change to robot-patient interaction code that is governed by, and clears, an automated verification process consistent with a predetermined change control plan under section 515C and the verification requirement of section 515D.*].”.

(g) Section 520(o) (21 U.S.C. 360j(o)); Clinical Decision Support.

“The term ‘device’ ... shall not include a software function ... [insert: *except that this paragraph does not exclude software that controls or generates the motion of a robot in direct physical contact with a patient, which remains a device subject to section 515D.*].”.

(h) Section 521 (21 U.S.C. 360k); State and Local Requirements.

“... no State or political subdivision of a State may establish ... any requirement [insert: *except that nothing in this section preempts a State requirement that preserves a licensed human’s authority over an artificial-intelligence-influenced decision and is not different from, or in addition to, a requirement under section 515D.*].”.

- (i) Remaining Conforming Amendments.** Sections 505F (21 U.S.C. 355g), 513 (21 U.S.C. 360c), 515 (21 U.S.C. 360e), and 515C (21 U.S.C. 360e-4) are amended as described in Table 3 and section 3(b), each by inserting the section-515D recognition, the graduated special controls, the cleared-record contents requirement, and the automated verification-before-change protocol, respectively. No existing text is struck by any conforming amendment; each change is an insertion, so existing law is preserved in full [28, 30, 31, 32].

APPENDICES: THE TWELVE SUBMISSION DELIVERABLES

The following twelve appendices are the submission-deliverables package, each a complete, submission-quality document converted to LaTeX and listed in the table of contents. They accompany the operative bill and never replace its text.

APPENDIX A. ONE-PAGE SUMMARY (NON-OPERATIVE).

Independent research draft. Bill: H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session. Refer to: House Committee on Energy and Commerce (Subcommittee on Health).

The problem: a sequencing gap that embodiment makes dangerous. Physical AI oncology trials place autonomous and semi-autonomous robots in direct physical contact with patients, and the robot’s behavior at the moment of contact is set by software that is increasingly machine-generated. No Federal law governs the order of operations, so a sponsor may generate robot-patient interaction code, run it, and document the checks afterward. In a disembodied data pipeline an error yields a wrong number a human can catch downstream; in an embodied surgical system the same error class yields a wrong motion no human can catch in time.

The solution: verify before generate, in a new section 515D. The bill adds a new section 515D to the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 360e-5), inserted right after the predetermined change control plan authority (section 515C, 21 U.S.C. 360e-4). No robot-patient interaction code may be generated or executed unless an automated verification, validation, and uncertainty quantification process, bound to named external standards, has first cleared, with the record documented and attested. Each gate emits ACCEPT, BLOCK, or ESCALATE; the verification fraction must equal 1.0; and three gates carry a hard catastrophe predicate at agreement 1.00.

What it changes: ten conforming amendments. Ten targeted edits thread the rule through Title 21: sections 201(h), 301, 501, 505F, 510(k), 513, 515, 515C, 520(o), and 521, so the new rule is enforceable and consistent with existing device law [25, 26, 27, 30, 31].

The assurance evidence. VVUQ-01 held the verification fraction at 1.0 across 51 of 51 tests, accepting 1 candidate and blocking 5, with 1 escalation. VVUQ-02 cleared all 10 gates over a surgical-humanoid control codebase: 172 of 172 tests, 32 of 32 sweep iterations, composite mean 93.56, reproducible from deterministic seed 20260525 [3, 4].

In-force analogues and cost. The approach builds on the section 515C change control plan authority and the FDA final PCCP guidance, the Quality Management System Regulation (21 CFR part 820, effective February 2, 2026), 21 CFR part 11, and the Medicare Advantage human-over-AI guardrail (42 CFR 422.101(c)). It directs final regulations within 365 days and creates no new spending program, so it is expected to be PAYGO neutral [5, 6, 15].

Table 5 condenses the one-page summary into a block table.

Block	In one line
Problem	No Federal rule makes verification precede generation of robot-patient code; embodiment turns a wrong number into a wrong motion
Rule	New section 515D: clear an automated, standards-bound VVUQ gate before generating or executing the code
Reach	Ten conforming amendments thread the rule through Title 21
Evidence	51 of 51 and 172 of 172 tests; ten gates; composite 93.56; seed 20260525
Cost	Rulemaking only; expected PAYGO neutral

Block	In one line
The ask	A House sponsor to introduce and refer to Energy and Commerce

Table 5. The one-page summary as a block table: the problem, the rule, its reach, the evidence, the cost, and the ask.

APPENDIX B. SECTION-BY-SECTION ANALYSIS (NON-OPERATIVE).

This appendix walks through H. R. 9510 provision by provision and gives a plain-English gloss of each part. It is an annotation, not operative text: where the gloss and the enacted words differ, the words of section 3 and the conforming amendments govern. The bill amends the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 301 et seq.), current through Public Law 119-93, and its operative mechanism is a single sequencing rule in a new section 515D (21 U.S.C. 360e-5), inserted after section 515C (21 U.S.C. 360e-4): an automated verification, validation, and uncertainty quantification process must clear robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation [37, 38, 32].

Sec. 1. Short Title; Table of Contents. Subsection (a) gives the citation, the “Verification Before Generation in Physical AI Oncology Trials Act of 2026.” Subsection (b) lays out the table of contents and maps what follows. This is standard front matter that fixes how the Act is cited and carries no substantive duty.

Sec. 2. Findings. Section 2 records fourteen congressional findings that justify the rule and show it is feasible in 2026. They state the problem (robots in direct physical contact with patients, behavior at contact governed by increasingly machine-generated software, no Federal rule that the code be verified before it is generated or executed); explain why embodiment raises the stakes (a disembodied error yields a wrong number a human can catch downstream, while an embodied error yields a wrong motion no human can catch in time); identify the closest existing analogues (the section 515C change control plan authority and FDA guidance, the Quality Management System Regulation at 21 CFR part 820 in force since February 2, 2026, and 21 CFR part 11), noting each is voluntary and not required before generation; and ground the mechanism in consensus standards, in human-over-AI precedent in State law and the Medicare Advantage guardrail, and in Federal nondiscrimination law. The findings supply the predicate a court or agency would consult when construing the Act [32, 6, 21].

Sec. 3. Amendment to the Federal Food, Drug, and Cosmetic Act. The operative heart of the bill. Subsection (a) inserts the new section 515D into subchapter V after section 515C; subsection (b) makes the ten conforming amendments; subsection (c) is the clerical amendment; subsection (d) is the rule of construction; and subsection (e) sets the effective date. Section 515D runs to subsections (a) through (j), glossed next.

Sec. 3, section 515D(a) Requirement. The central rule. No robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated verification, validation, and uncertainty quantification process, bound to one or more named external standards, has first cleared for that interaction and the cleared record has been documented and attested under subsection (e). This converts “verify before generate” from an engineering best practice into a binding statutory precondition.

Sec. 3, section 515D(b) Order of Operations. Locks the sequence. A sponsor may not generate, and may not execute, the code in advance of the cleared verification record, and a record created afterward does not satisfy the section. This forecloses generating and running code first and documenting later; the timing of the record is itself the compliance fact.

Sec. 3, section 515D(c) Gate Thresholds and Decision Rule. The quantitative test each interaction must pass. Every gate emits ACCEPT, BLOCK, or ESCALATE: a single failing dimension sets

BLOCK, and ambiguity or divergence sets ESCALATE and returns the interaction to a qualified human. For each interaction the verification fraction must equal 1.0, the validation agreement must meet or exceed the per-gate threshold with the maximum relative error at or below the per-gate bound and the human review recorded, the coefficient of variation across not fewer than three seeded runs must stay within the per-gate bound, and any applicable hard catastrophe predicate must also hold. The ten-gate schedule binds each gate to a minimum agreement, a maximum coefficient of variation, the presence of a hard predicate, and a governing external standard, making the gate auditable rather than discretionary; gates 06, 09, and 10 carry the hard predicate at agreement 1.00 [7, 39].

Sec. 3, section 515D(d) Readiness Gates. Site-level and robot-level preconditions before a patient is enrolled or a robot acts. The trial site must pass the Physical AI Standard Level gate on legislative authorization, regulatory compliance, and operational readiness. Each robot must meet the minimum Unification Standard Level for its functional type: not less than 7.0 for a surgical type, 6.0 for a therapeutic positioning or diagnostic needle placement type, 4.0 for a rehabilitative type, and 3.0 for a companion monitoring type. And each robot must be validated across not fewer than two simulation frameworks within cross-framework tolerances of less than 2 millimeters of trajectory deviation and less than 0.5 newton of force discrepancy. The graduated floor matches scrutiny to the physical risk of the function.

Sec. 3, section 515D(e) Documentation and Attestation. The record subsection (a) requires. For each covered algorithm the sponsor files algorithm documentation: a plain-language summary, the decision logic or architecture, the training or conditioning data sets, the gate bindings identifying the governing standard and threshold, the verification-before-generation record, and a determinism statement giving the seed. A responsible officer signs a compliance attestation certifying that the gate cleared before generation, that each gate met its standard-bound threshold, that the site and robots cleared their readiness gates, that bias on protected characteristics is minimized, and that a licensed human retained the medical-necessity decision. All records sit in a hash-chained, tamper-evident audit trail satisfying 21 CFR part 11, making the duty enforceable and the attestation a personal certification [21].

Sec. 3, section 515D(f) Cybersecurity, Human Oversight, and Lifecycle. Operational controls across the system's life. Each Physical AI system must incorporate authentication and access control, encryption in transit and at rest, network segmentation, and software-integrity verification; operate under recorded human oversight with a manual override and an emergency stop that reaches a safe state within a procedure-appropriate budget plus hand-back-to-human escalation; and manage configuration and model versions, monitor for model and data drift, and, on decommissioning, archive the audit trail, securely erase patient data, and revoke access. These duties guard the gate against tampering and degradation over time.

Sec. 3, section 515D(g) Nondiscrimination. Requires that each covered algorithm and its training data minimize the risk of bias on protected characteristics and adhere to evidence-based clinical guidelines, and that each system be designed and operated consistent with applicable accessibility and accommodation requirements. This carries Federal civil-rights and accessibility norms into the gate as substantive criteria, not afterthoughts.

Sec. 3, section 515D(h) Regulations. Directs the Secretary to issue final regulations not later than 365 days after enactment, and permits interim guidance in the meantime, with a sponsor able to comply by meeting subsections (a) through (g) until the regulations issue. This makes the statute self-executing rather than dormant pending rulemaking.

Sec. 3, section 515D(i) Definitions. Defines the fifteen operative terms, including Physical AI system, robot agent, robot-patient interaction code, generative artificial intelligence, AI-enabled device software function, verification, validation, uncertainty quantification, verification fraction (an equality to 1.0), validation agreement, coefficient-of-variation bound (over not fewer than three seeded runs), hard catastrophe predicate, Physical AI Standard Level, Unification Standard Level (a 1.0 to 10.0 robot-readiness score), and hash-chained audit trail. Precise definitions are what let the quantitative gate operate without discretion.

Sec. 3, section 515D(j) Rule of Construction. Confirms the section's limits. It does not displace the investigational-device exemption under section 520(g), human-subject protection under 21 CFR part 50, the IND framework under 21 CFR part 312, or the section 515C change-control authority as it applies outside Physical AI, and it reclassifies no device. This positions section 515D as an additive sequencing gate layered on the existing trial framework, not a replacement for it [35, 36].

Sec. 3(b) Conforming Amendments (1) through (5). Subsection (b) threads the new duty through ten device provisions so the Act stays internally consistent. Amendment (1), section 201(h) (21 U.S.C. 321(h)), confirms that software that generates or executes robot-patient interaction code, or that directs the autonomous physical action of a robot in contact with a human subject, is a device, pulling the regulated code inside FDA jurisdiction. Amendment (2), section 301 (21 U.S.C. 331), adds a prohibited act reaching generation or execution of the code in violation of section 515D and the failure to make or file the required documentation and attestation, the enforcement hook. Amendment (3), section 501 (21 U.S.C. 351), deems a Physical AI device software function used without a cleared section 515D record adulterated. Amendment (4), section 505F (21 U.S.C. 355g), recognizes section 515D records as real world evidence and applies the program to devices in Physical AI oncology investigations. Amendment (5), section 510(k) (21 U.S.C. 360(k)), provides that a verification-governed change consistent with an established predetermined change control plan needs no new premarket notification [25, 26, 27, 28, 29].

Sec. 3(b) Conforming Amendments (6) through (10). Amendment (6), section 513 (21 U.S.C. 360c), graduates special controls by degree of autonomy (assistive, augmentative, or autonomous), matching rigor to how much the system acts on its own. Amendment (7), section 515 (21 U.S.C. 360e), requires a premarket approval application for such a device to include the cleared verification record and gate evidence. Amendment (8), section 515C (21 U.S.C. 360e-4), requires a predetermined change control plan for such a device to include an automated verification-before-change protocol consistent with section 515D, carrying the rule forward into the lifecycle. Amendment (9), section 520(o) (21 U.S.C. 360j(o)), confirms that such a software function is not within the clinical-decision-support exclusion and remains a device, closing the carve-out. Amendment (10), section 521 (21 U.S.C. 360k), adds a savings clause preserving State human-in-the-loop and audit requirements that are not different from, or in addition to, a Federal device requirement, protecting the State human-over-AI laws cited in the findings [30, 31, 32, 33, 34].

Sec. 3(c) Clerical Amendment. Inserts the new “Sec. 515D” item into the Act’s table of contents after the item for section 515C, so the codified table reflects the new section.

Sec. 3(d) Rule of Construction. Restates at the Act level that nothing in the Act displaces the section 520(g) investigational-device exemption, 21 CFR part 50 human-subject protection, the 21 CFR part 312 IND framework, the section 515C authority outside Physical AI, or any State law preserved under section 521 as amended, and that nothing reclassifies any device. It guards against reading the bill more broadly than intended.

Sec. 3(e) Effective Date; Implementation Deadline; Transition Rule. The Act takes effect on the first day of the first calendar quarter beginning not less than one year after enactment; the Secretary

must issue the final section 515D(h) regulations not later than 365 days after enactment; and an investigation already enrolling on the effective date has 180 days to come into compliance, with the February 2, 2026, effective date of the Quality Management System Regulation left undisturbed. This gives sponsors and the agency a predictable runway [6].

Sec. 4. Comparative Print; Changes in Existing Law. Section 4 reproduces the eleven affected Title 21 device sections in 21 U.S.C. order, showing only the affected provisions, with matter to be deleted struck and matter to be inserted marked. It is a Ramseyer-style comparative print, reproduced in full at Appendix E: it carries no new duty but lets a reader see exactly how each conforming amendment changes existing law, the transparency record a committee expects before introduction.

APPENDIX C. PLAIN-ENGLISH POLICY MEMORANDUM (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress. This memorandum is non-operative; it explains the bill and carries no legal force of its own.

TO: Health policy staff, Member of the House Committee on Energy and Commerce (Subcommittee on Health)

FROM: CEO Kevin Kawchak, ChemicalQDevice

RE: H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session

DATE: June 7, 2026

This memorandum lays out one specific safety gap in how artificial intelligence is now used in cancer trials, explains why current law does not close it, and describes a narrow, low-cost bill that would. It is written so a non-engineer can read it in a few minutes and brief the Member, and it is set out as a short series of questions and answers.

Q. What is the problem in one sentence?

A. Cancer trials are now placing robots in direct physical contact with patients, what the robot does at the instant it touches a patient is decided by software that AI increasingly writes in real time, and no Federal law governs the order of operations, so a sponsor may generate the robot-patient interaction code, run it on a patient, and document the safety checks afterward. We call this the sequencing gap.

Q. Why does the order of operations matter so much here?

A. Because of embodiment. In an ordinary data pipeline a software error produces a wrong number a human can catch downstream before anyone is harmed; in an embodied surgical system the same class of error produces a wrong physical motion a human cannot catch in time, because one moving body concentrates every failure mode into a single instrument acting in real time. Checking the code after it has already moved is, in this setting, checking too late.

Q. Does not existing law already cover this?

A. The framework is substantial, and the bill builds on it rather than replacing it, but every piece stops just short of requiring verification before the code is generated. The closest analogue is the pre-determined change control plan authority Congress added in 2022 at section 515C of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 360e-4), with the FDA final guidance for AI-enabled device software functions. That authority lets the FDA pre-authorize a planned change before it ships, which is the right idea, but it is voluntary, submission-based, runs on the FDA review clock, and is not required in advance of generating new code. The Quality Management System Regulation

(21 CFR part 820, in force since February 2, 2026) and the part 11 electronic-records rule make a verification record trustworthy but do not require the verification to come first; the CMS Medicare Advantage guardrail (42 CFR 422.101(c)) keeps a human over AI for a coverage decision but not over robot code; and Federal nondiscrimination law reaches bias and accessibility but not sequencing [32, 5, 6, 15].

- A. Several States have acted, but at the claims and utilization-review layer, not the robot-code layer: California Senate Bill 1120, Illinois House Bill 1806, the Texas Responsible Artificial Intelligence Governance Act (House Bill 149), and Maryland House Bill 820. No two are identical, and none reaches trial-conduct code, so a sponsor operating across States cannot build one robot-patient code base against divergent claims-layer regimes. The substantive content a verification gate would enforce already exists, scattered across these authorities; the one rule that makes verification a precondition of generation does not exist anywhere, and that single missing rule is what H. R. 9510 supplies.

Q. What exactly does the bill do?

- A. It amends the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 301 et seq.), current through Public Law 119-93, adding one new section, 515D, codified at 21 U.S.C. 360e-5, placed immediately after the section 515C change control authority it most resembles. The core rule is a single sequencing requirement: no robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated verification, validation, and uncertainty quantification process, bound to named external standards, has first cleared for that interaction, with the record documented and attested. A record created after generation or execution does not satisfy the section; the timing of the record is itself the compliance fact. Each check emits ACCEPT, BLOCK, or ESCALATE; the verification fraction must equal 1.0, with no partial credit; and three gates carry a hard catastrophe predicate at perfect agreement, tied to ASME V&V 40, ISO/TS 15066, and the other named standards so the test is auditable rather than discretionary [37, 7, 10].

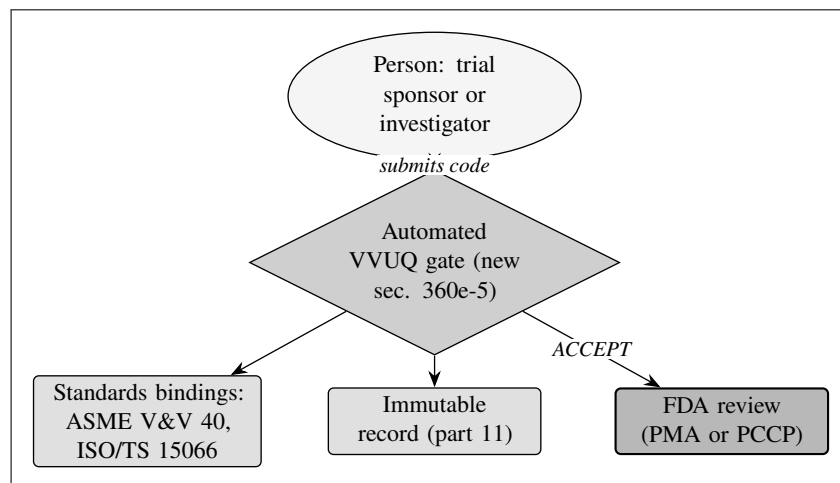


Figure 8. The verification gate as a system: a submitted interaction is checked against named standards, written to an immutable record, and only an ACCEPT artifact proceeds to FDA review.

Figure 8 shows the gate as a working system rather than a slogan. A person, the sponsor or investigator, submits the robot-patient interaction code; the automated gate under new section 515D checks it against the named external standards and writes the result to an immutable part 11 record; and only an ACCEPT artifact is allowed to proceed to FDA review under a premarket application or a predetermined change control plan. A BLOCK or an ESCALATE never reaches that arrow.

Q. Is it actually feasible to verify code before it runs?

- A. Yes, and it has already been run on the record, not merely hoped for. In a first study (VVUQ-01) an automated pipeline held the verification fraction at 1.0 across 51 of 51 tests, accepting 1 candidate artifact, blocking 5, and escalating 1. In a second study (VVUQ-02) a ten-gate assurance suite over the control software for a surgical humanoid cleared 172 of 172 tests and 32 of 32 sweep iterations, with a composite mean of 93.56, reproducibly from a fixed seed (20260525). The bill asks sponsors to do, as a precondition, what has already been demonstrated to work [3, 4].

Q. Why now, and not in a few years?

- A. Clinical investigation is moving toward real-time, decentralized, machine-generated control code. A real-time trial cannot wait weeks for a manual, document-by-document code review between procedures, and manual review already cannot keep pace with the volume of code a multi-site trial produces in 2026. The gap widens with every robot added to a site, so an automated gate that clears before generation is not a brake on modernization; it is the only assurance that can run at the speed modernization now demands.

Q. How does the bill fit existing law, and what does it cost?

- A. It is additive, not disruptive. It leaves the investigational-device exemption, human-subject protection (21 CFR part 50), and the investigational new drug framework (21 CFR part 312) in place; frames the gate as a pre-authorized change control discipline of the kind section 515C already contemplates; and routes its records through the part 11 rule and the part 820 quality system that 2026 sponsors already meet. A savings clause preserves the State human-in-the-loop laws above. It creates no new agency or program, directs final regulations within 365 days, and is expected to be PAYGO neutral, with the statute self-executing in the meantime [35, 36, 40].

Q. What is the recommendation?

- A. We recommend that the Member consider introducing H. R. 9510 and refer it to the Committee on Energy and Commerce (Subcommittee on Health). It closes a narrow, well-defined, and growing safety gap with a proven, low-cost mechanism; it builds on tools the FDA and CMS already use; it preserves existing State protections; and it positions the United States to set the global standard for verifying surgical-robot software before that software ever touches a patient.

APPENDIX D. LEGISLATIVE FINDINGS (NON-OPERATIVE).

This appendix reproduces the legislative findings of fact set out in section 2. The findings are not operative commands; they supply the evidentiary and legal predicate a court or agency would consult when construing the Act. Congress finds the following.

- (1) Physical AI oncology clinical investigations place autonomous and semi-autonomous robots in direct physical contact with patients, and the behavior of such a robot at the moment of contact is determined by software that is increasingly machine-generated.
- (2) No Federal law requires the verification of robot-patient interaction code to clear before that code is generated or executed; a sponsor may generate, execute, and document afterward, because the order of operations is not regulated.
- (3) Embodiment raises the stakes from software quality to patient safety: a disembodied error yields a wrong number a human can catch downstream, while an embodied error yields a wrong motion a human cannot catch in time.

- (4) The predetermined change control plan authority in section 515C (21 U.S.C. 360e-4) and the FDA final guidance on such plans for AI-enabled device software functions are the closest existing analogues, but each remains voluntary, submission-based, and neither automated nor required in advance of generation.
- (5) The Quality Management System Regulation (21 CFR part 820, effective February 2, 2026) and the electronic-records requirements of 21 CFR part 11 supply quality and record-integrity scaffolding but impose no rule that verification precede generation.
- (6) ASME V&V 40-2018 and the FDA computational-modeling credibility guidance establish a credibility framework, in which model risk is the product of model influence and decision consequence, that an automated gate can implement.
- (7) NASA-STD-7009A, IEEE Std 7009-2024, IEC 80601-2-77, ISO 14971, and ISO/TS 15066 supply the dispersion bounds, fail-safe requirements, and biomechanical contact limits to which such a gate may be bound.
- (8) Automated verification is feasible today: a pipeline held the verification fraction at 1.0, accepting one candidate and blocking five, and a ten-gate assurance suite over a surgical humanoid cleared reproducibly from a fixed seed.
- (9) Human-over-AI review is widely adopted in State law, including California Senate Bill 1120, Illinois House Bill 1806, Texas House Bill 149, and Maryland House Bill 820, each preserving a licensed human's authority.
- (10) The strongest Federal precedent for mandatory human-over-AI oversight is the Medicare Advantage guardrail (42 CFR 422.101(c)), under which an algorithm alone may not justify a coverage denial.
- (11) Federal nondiscrimination law, including 45 CFR 92.210, title VI (42 U.S.C. 2000d), and the Americans with Disabilities Act (42 U.S.C. 12101 et seq.), requires that a patient-facing algorithm minimize bias and respect accessibility.
- (12) The American Medical Association artificial-intelligence taxonomy (assistive, augmentative, autonomous) provides a graduated spine for matching verification rigor to the degree of autonomy of a device.
- (13) A Physical AI system is at once a medical product and a research tool, and the requirement is agnostic to that classification, because its patient-facing code must clear the gate before it runs.
- (14) As clinical investigation moves toward real-time and decentralized designs under harmonized good clinical practice, manual review cannot keep pace, and an automated ex ante gate serves rather than impedes that modernization.

Table 6 maps each finding to the principal authority on which it rests. Statutes, in-force rules, and recognized consensus standards carry operative weight; demonstrated engineering results establish feasibility.

Finding	Principal authority
(1), (3)	National Physical AI platform record; VVUQ-01 method and pipeline
(2)	Absence of a verify-before-generate mandate in Title 21
(4)	FDCA section 515C (21 U.S.C. 360e-4); FDA final PCCP guidance
(5)	Quality Management System Regulation (21 CFR part 820); 21 CFR part 11
(6)	ASME V&V 40-2018; FDA credibility guidance

Finding	Principal authority
(7)	NASA-STD-7009A; IEEE 7009-2024; IEC 80601-2-77; ISO 14971; ISO/TS 15066
(8)	VVUQ-01 (fraction 1.0; 1 accept, 5 block); VVUQ-02 (ten-gate suite, seed)
(9)	CA SB 1120; IL HB 1806; TX HB 149; MD HB 820
(10)	Medicare Advantage AI guardrail (42 CFR 422.101(c))
(11)	45 CFR 92.210; title VI (42 U.S.C. 2000d); ADA (42 U.S.C. 12101 et seq.)
(12)	AMA artificial-intelligence taxonomy
(13), (14)	National Physical AI platform record; ICH E6(R3)

Table 6. The fourteen findings mapped to their principal authorities, for auditability. Only enacted statutes, in-force rules, and recognized standards carry operative weight.

APPENDIX E. RAMSEYER COMPARATIVE PRINT (NON-OPERATIVE).

Reader’s note on marking. This comparative print shows the changes H. R. 9510 makes to existing law, reproduced from the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 301 et seq.) as carried in Title 21, United States Code, current through Public Law 119-93. Existing law in which no change is proposed is shown in plain text. Matter proposed to be inserted is shown [insert: *like this insert marker*]; matter proposed to be deleted is shown [strike: *like this strike marker*], with the marker text written out literally, so a reader need not rely on strikethrough or underline rendering. Unchanged matter omitted for brevity is shown by a row of asterisks. The print reaches the eleven affected Title 21 device sections, reproduced in 21 U.S.C. order and shown only as to their affected provisions, consistent with section 4 of the Act. It carries no new duty; it is the transparency record a committee expects before introduction [38, 37].

Section 301 (21 U.S.C. 331 note); Short Title. A new statutory note is added after the existing short-title notes of the Act:

Short Title of 2026 Amendment. [insert: *Pub. L. 119-XXX, section 1, approved in 2026, provided that: “This Act may be cited as the ‘Verification Before Generation in Physical AI Oncology Trials Act of 2026’.”*]

Section 201(h) (21 U.S.C. 321(h)); Definition of Device. The device definition is amended by adding at the end of paragraph (1):

The term “device” ... does not include software functions excluded pursuant to section 360j(o) of this title. [insert: *A software function that generates or executes robot-patient interaction code (as defined in section 515D), or that directs the autonomous physical action of a robot in contact with a human subject in a clinical investigation, is a device and is not a software function excluded under section 360j(o).*]

Section 301 (21 U.S.C. 331); Prohibited Acts. A new prohibited act is added after subsection (iii):

(iii) The refusal or failure to follow an order under section 364g of this title. [insert: *(jjj) The generation or execution of robot-patient interaction code in a Physical AI oncology clinical investigation in violation of section 515D, or the failure to make or file the documentation and attestation required by section 515D before such generation or execution.*]

Section 501 (21 U.S.C. 351); Adulterated Devices. A new paragraph is added after subsection (j):

(j) Delayed, denied, or limited inspection ... refuses to permit entry or inspection. [insert: *(k) Physical AI device software function used without a cleared verification record. If it is a Physical AI device software function (as defined in section 515D) used to generate or execute robot-patient interaction code in a clinical investigation without a cleared verification record required by section 515D, or in a manner not in conformity with a predetermined change control plan governed by section 515D.*]

Section 505F (21 U.S.C. 355g); Utilizing Real World Evidence. Subsection (b) is amended, and a new subsection (g) is added:

(b) Real world evidence defined. ... means data regarding the usage, or the potential benefits or risks, of a drug derived from sources other than traditional clinical trials. [insert: *Such term includes verification, validation, and uncertainty quantification records and trial telemetry generated under section 515D.*]

[insert: (g) *Application to Physical AI oncology investigations. The Secretary shall, in carrying out the program under this section, evaluate the use of real world evidence, including records generated under section 515D, to support the evaluation of devices used in Physical AI oncology clinical investigations.*]

Section 510(k) (21 U.S.C. 360(k)); Premarket Notification. The subsection is amended by adding at the end:

... Such certification shall not be considered an element of such notification. [insert: *A change to a device that generates or executes robot-patient interaction code, cleared under this subsection, that is consistent with a verification protocol under section 515D and an established predetermined change control plan under section 360e-4, shall not require a new premarket notification under this subsection.*]

Section 513 (21 U.S.C. 360c); Classification of Devices. A new subsection (l) is added:

[insert: (l) *Physical AI device software functions. In classifying a device that generates or executes robot-patient interaction code, the Secretary shall consider the degree of autonomy of the device (assistive, augmentative, or autonomous) and may establish special controls, including a verification protocol under section 515D, commensurate with that degree of autonomy. A device of autonomous degree used in a Physical AI oncology clinical investigation shall be subject to the verification requirements of section 515D regardless of class.*]

Section 515 (21 U.S.C. 360e); Premarket Approval. The contents of a premarket approval application are amended by striking the conjunction at the end of subparagraph (G), revising subparagraph (H), and adding new subparagraph (I):

(G) the certification required under section 282(j)(5)(B) of title 42 (which shall not be considered an element of such application)[strike: ; and]

(H) such other information relevant to the subject matter of the application as the Secretary, with the concurrence of the appropriate panel under section 360c of this title, may require[strike: .][insert: ; and]

[insert: (I) *in the case of a device that generates or executes robot-patient interaction code, the cleared verification record and the gate evidence required by section 515D.*]

Section 515C (21 U.S.C. 360e-4); Predetermined Change Control Plans. The scope paragraphs for approved and cleared devices are amended, and a new subsection (d) is added. In subsection (a)(3) (approved devices), and identically in subsection (b)(3) (cleared devices), the scope clause is amended by inserting before the period:

... and performance requirements for changes made under the plan[insert: , and, for a device that generates or executes robot-patient interaction code, an automated verification-before-change protocol consistent with section 515D that must clear before any change is implemented].

[insert: (d) *Physical AI device software functions. A predetermined change control plan for a device that generates or executes robot-patient interaction code shall require that each planned change clear the verification process required by section 515D, in advance of generation or execution, as a condition of being implemented without a new submission under section 360e or section 360(k).*]

Section 520(o) (21 U.S.C. 360j(o)); Clinical Decision Support Software. Paragraph (1) is amended by adding at the end:

... so that it is not the intent that such health care professional rely primarily on any of such recommendations to make a clinical diagnosis or treatment decision regarding an individual patient. [insert: *The exclusion in this paragraph does not apply to, and the term “device” includes, a software function that generates or executes robot-patient interaction code, or that directs the autonomous physical action of a robot in contact with a human subject in a clinical investigation; such a function is a device subject to section 515D.*]

Section 521 (21 U.S.C. 360k); State and Local Requirements. A new subsection (c) is added:

[insert: (c) *Rule of construction for Physical AI oversight. Nothing in this section preempts a State requirement of general applicability that a licensed human retain the authority to monitor, intervene, override, or terminate an artificial intelligence or robotic system, or that an audit record be maintained, to*

the extent that the requirement is not different from, or in addition to, a requirement applicable under this chapter to a device, including the verification requirements of section 515D.]

New Section 515D (21 U.S.C. 360e-5); Verification Before Generation. The following new section is inserted in subchapter V immediately after section 515C (21 U.S.C. 360e-4). It is added in its entirety; the full text of subsections (a) through (j) is set out in section 3 of this Act:

[insert: Sec. 515D. Verification before generation of robot-patient interaction code in Physical AI oncology investigations. (a) Requirement. No robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated verification, validation, and uncertainty quantification process, bound to one or more named external standards, has first cleared for that interaction, and the cleared verification record has been documented and attested under subsection (e). (b) through (j) Order of operations, gate thresholds and decision rule, readiness gates, documentation and attestation, cybersecurity, human oversight, and lifecycle, nondiscrimination, regulations, definitions, and rule of construction, are set out in full in section 3 of this Act.]

Each change above is an insertion except the two short strikes in section 515 (the conjunction at the end of subparagraph (G) and the period at the end of subparagraph (H)), so existing law is otherwise preserved in full and the new rule is threaded consistently through Title 21 [31, 32].

APPENDIX F. CONSTITUTIONAL AUTHORITY STATEMENT (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress.

Under clause 7(c) of Rule XII of the Rules of the House of Representatives, a bill or joint resolution may not be introduced unless the sponsor submits, for printing in the Congressional Record, a statement citing as specifically as practicable the power or powers granted to Congress in the Constitution to enact the measure. For H. R. 9510, which amends the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 301 et seq.), the operative statement is the single sentence below [41, 37].

Congress has the power to enact this legislation pursuant to the following: Article I, section 8, clause 3, of the Constitution of the United States (the Commerce Clause), which empowers Congress to regulate commerce among the several States, on which the Federal Food, Drug, and Cosmetic Act has rested since its enactment in 1938, as supplemented by Article I, section 8, clause 18 (the Necessary and Proper Clause).

The paragraphs that follow explain and support that sentence; they are not part of the statement printed under clause 7(c).

Why the Commerce Clause. The Federal Food, Drug, and Cosmetic Act has rested on the Commerce Clause since 1938, and the Act's prohibited-acts provisions (21 U.S.C. 331) are written expressly in terms of articles introduced into, or held for sale after shipment in, interstate commerce. Because H. R. 9510 amends that Act, by adding a new section 515D (codified at 21 U.S.C. 360e-5) and conforming amendments threaded through Title 21, it draws on the same constitutional source as the statute it modifies. The bill adds no new jurisdictional theory; it extends a recognized one to a new class of regulated articles [26, 25].

Devices and their code are articles of interstate commerce. Surgical and oncology robots, their components, and the software that governs robot-patient interaction are manufactured, marketed, shipped, and used across State lines, and the device authorities the bill builds on, classification under 21 U.S.C. 360c, premarket review under 21 U.S.C. 360e and section 360(k), and predetermined change control plans under 21 U.S.C. 360e-4, all regulate that interstate market. The robot-patient

interaction code subjected to verification before generation is itself a traded, updatable article: it is distributed to clinical-investigation sites in multiple States, modified through interstate software pipelines, and integrated into devices that move in commerce. Such code and the Physical AI systems that execute it are both instrumentalities and articles of interstate commerce, squarely within the reach of clause 3 [30, 31, 34, 32].

Aggregate effects and the Necessary and Proper Clause. The bill also reaches activity that, in the aggregate, substantially affects interstate commerce: Physical AI oncology clinical investigations are conducted by sponsors and sites operating in a national research market, generating data, evidence, and devices that enter interstate channels and inform nationwide marketing and Medicare coverage determinations. Setting a uniform sequencing rule, that automated verification, validation, and uncertainty quantification must clear before robot-patient code is generated or executed, is a regulation of the terms on which these articles and services may be placed in that market. The Necessary and Proper Clause supports the means chosen: the documentation, attestation, audit-trail, cybersecurity, and human-oversight duties in section 515D are appropriate and plainly adapted to making the underlying device and commerce regulation effective.

Filing. In practice the statement accompanies the measure at the point of introduction rather than as a separate later filing. The House has modernized intake through the electronic hopper, and a sponsoring office can generate and submit the constitutional authority statement, together with the measure, within that system; the Clerk then makes each statement available to the public in electronic form. The table below summarizes the requirement [42, 41].

Item	Detail
Governing rule	Rule XII, clause 7(c), Rules of the House of Representatives
What is filed	A statement citing as specifically as practicable the constitutional power or powers to enact the measure
Primary citation	Article I, §8, clause 3 (Commerce Clause)
Supporting citation	Article I, §8, clause 18 (Necessary and Proper Clause)
Where it appears	Printed in the Congressional Record; made available electronically by the Clerk
How it is submitted	With the measure through the electronic hopper at introduction

The constitutional authority requirement of Rule XII, clause 7(c): the primary citation is the Commerce Clause, with the Necessary and Proper Clause in support, filed with the measure at introduction.

APPENDIX G. PAYGO AND COST ESTIMATE NOTE (NON-OPERATIVE).

Independent research draft. This is an informal, pre-introduction estimate, not an official Congressional Budget Office (CBO) cost estimate or score, and is not endorsed by the FDA, HHS, the CBO, or any Member of Congress. Bill: H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session. Refer to: House Committee on Energy and Commerce (Subcommittee on Health).

Summary. H. R. 9510 sets one sequencing rule for Physical AI oncology clinical investigations: an automated verification, validation, and uncertainty quantification process must clear robot-patient interaction code before that code is generated or executed. The bill operates through the existing device authorities of the Federal Food, Drug, and Cosmetic Act, adding a new section 515D (21 U.S.C. 360e-5) together with conforming amendments. It creates no entitlement, no direct spending program, no new tax, and no mandatory appropriation. Its budgetary effects are discretionary: the Food and Drug Administration must issue final regulations under section 515D and then fold a new sequencing check into existing review and inspection workflows. Those costs are expected to be absorbable within existing FDA device-program resources and the medical-device user-fee

structure. On that basis the bill is expected to be neutral under the Statutory Pay-As-You-Go Act of 2010, which reaches direct spending and revenue rather than discretionary agency activity [40, 32].

Basis of the estimate. This note rests on the text of H. R. 9510 and on how its mechanism maps onto existing FDA authorities. The operative provision, new section 515D, regulates the order of operations within investigations already conducted under the device framework: classification under 21 U.S.C. 360c, premarket review under 21 U.S.C. 360e and section 510(k) (21 U.S.C. 360(k)), and predetermined change control plans under section 515C (21 U.S.C. 360e-4). The bill adds no program office, grant authority, benefit, or collection that would alter Federal receipts. Its cost is therefore the incremental agency effort to write one rulemaking and to embed a sequencing check in existing review and inspection workflows. Because no formal CBO estimate has issued, the assessments here are qualitative and conservative; informal consultation with CBO can precede introduction, and a formal score typically follows [30, 31, 29].

Direct spending and revenue (PAYGO). Statutory PAYGO applies to legislation affecting direct spending or revenue. H. R. 9510 does not establish or expand any entitlement, mandatory grant, or other form of direct spending; its duties are conditions on conduct, enforced through the existing prohibited-acts and adulteration provisions of the Act (21 U.S.C. 331 and 21 U.S.C. 351) rather than through new outlays. The bill imposes no tax, levies no new user fee by statute, and changes no existing fee schedule, so it has no revenue effect. Any FDA implementation funded through appropriated user-fee authority is treated as discretionary spending, not direct spending, and is governed by appropriations rather than by the PAYGO scorecard. The expected scorecard entry is therefore zero, and the bill is expected to be PAYGO neutral [26, 27].

Discretionary cost (rulemaking and implementation). The principal cost driver is the direction in section 515D that the Secretary issue final regulations not later than 365 days after enactment, with interim guidance permitted in the meantime. Rulemaking of this scope, defining the binding standards, the gate semantics (ACCEPT, BLOCK, ESCALATE), the verification-fraction and validation-agreement floors, the documentation and attestation content, and the readiness gate, involves notice-and-comment drafting, economic and paperwork analysis, and interagency review. Once the rule takes effect, reviewers must read cleared verification records, confirm readiness-gate determinations, and assess attestations during investigational and premarket review, and inspectors must check sequencing compliance. These are extensions of work the device program already performs on software, change control plans, and quality systems under the Quality Management System Regulation (21 CFR part 820) and 21 CFR part 11, not a new function. The expected discretionary cost is modest and absorbable within existing FDA device-program resources and the medical-device user-fee structure, subject to confirmation in the formal CBO estimate and the appropriations process [6, 43, 21].

Unfunded mandates. Under the Unfunded Mandates Reform Act of 1995, the bill is not expected to impose an intergovernmental mandate above the statutory threshold; it regulates device sponsors and investigators rather than directing State, local, or tribal governments, and the savings clause of the Act (21 U.S.C. 360k) is preserved. The bill does impose a private-sector mandate on sponsors of Physical AI oncology clinical investigations, who must run the verification process before generating or executing robot-patient interaction code and must document and attest to the result. The expected cost is the incremental documentation, review, and attestation burden layered on validation activity that responsible sponsors already perform, so the private-sector mandate cost is expected to fall below the threshold, subject to CBO's formal determination [34].

Earmark declaration (negative). H. R. 9510 contains no congressional earmarks, no limited tax benefits, and no limited tariff benefits as those terms are defined in clause 9 of Rule XXI of the Rules

of the House of Representatives. This is a negative declaration: the bill makes no targeted appropriation, revenue, or tariff provision benefiting a specific entity or limited group. Its operative provisions are generally applicable regulatory requirements administered by the FDA, and the bill carries no appropriation.

Next step. This note is a good-faith pre-introduction estimate prepared to accompany H. R. 9510, not an official CBO product. The authoritative cost estimate, covering any direct spending and revenue effects, the discretionary spending projection over the standard budget window, and the formal mandate determination, is produced by the Congressional Budget Office and typically follows introduction and referral to House Energy and Commerce, with informal CBO consultation available beforehand. Nothing here should be read to bind, anticipate, or substitute for CBO's conclusions [40, 44].

Table 7 sets out the line items of this pre-introduction estimate.

Line item	Expected effect
Direct spending (mandatory)	None significant; no entitlement or mandatory program created
Revenue	None; no new tax, fee mandate, or revenue change
Statutory PAYGO scorecard	Neutral; no significant direct spending or revenue effect
Discretionary cost (rulemaking and implementation)	Modest; expected absorbable within existing FDA device-program and user-fee resources
Intergovernmental mandate (UMRA)	None expected above the statutory threshold
Private-sector mandate (UMRA)	A mandate exists; expected documentation and review costs are likely below the threshold
Earmarks, limited tax benefits, limited tariff benefits	None (negative declaration)
Formal CBO score	Follows introduction and referral to House Energy and Commerce

Table 7. PAYGO and cost line items for H. R. 9510. The bill directs rulemaking only, creates no direct spending or revenue, and is expected to be PAYGO neutral; the formal score is produced by the CBO after introduction.

APPENDIX H. SPONSOR AND COSPONSOR PACKET (NON-OPERATIVE).

Independent research draft. Not pending legislation and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress. No Member has agreed to sponsor this measure. Bill: H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session. Refer to: House Committee on Energy and Commerce (Subcommittee on Health). Governing rule: Rule XII of the Rules of the House of Representatives.

Purpose of this packet. This packet is the recruitment and sign-on instrument for the draft bill. In one place it tells a prospective sponsoring office what role the bill needs filled, what each signer is being asked to do, where the measure is expected to be referred, and what must be in hand before introduction. It does not introduce anything. Only a sitting Member of the House can introduce a House bill; an outside research project cannot file it directly. H. R. 9510 is illustrative, and the objective is singular: place the finished text in the hands of one Member who will introduce it and refer it to the Committee on Energy and Commerce, and gather original cosponsors to accompany it [41, 44].

The sponsor. A House bill has exactly one sponsor: the Member who introduces it and whose name leads the bill. Every other supporter is a cosponsor. The sponsor's office owns the introduced text and is the contact for the Clerk, the Parliamentarian, and the committee of referral. Two items are required from the sponsor before introduction. First, the sponsor's signature on the bill, which under current cloakroom guidance is required on the measure itself. Second, a signed Constitutional

Authority Statement: under Rule XII, clause 7(c)(1), a bill may not be introduced unless the sponsor submits a statement citing, as specifically as practicable, the constitutional power to enact it, here Article I, Section 8, Clause 3 (the Commerce Clause) on which the Act rests, supported by Clause 18 (the Necessary and Proper Clause). The best-fit sponsor is a Member of the Committee on Energy and Commerce, ideally on the Subcommittee on Health, with a demonstrated interest in medical-device safety, oncology care, or trustworthy AI; that office is also expected to route the final text through the House Office of the Legislative Counsel for proper form before filing [41, 45].

Original cosponsors. Original cosponsors are Members listed at the time of introduction, distinct from cosponsors added afterward. A public bill may have an unlimited number of cosponsors, and under Rule XII the sponsor names them. Bipartisan original cosponsors meaningfully improve a bill's prospects, so recruiting at least one Member from each party is a deliberate goal. To be counted as original, a Member must commit before or at introduction; once the bill is filed the original-cosponsor list closes and later supporters are added through the eHopper. Circulate for sign-on only after the sponsor approves the exact introduction version, so every signer endorses identical text. A balanced slate spans three interests on both sides of the aisle: device and FDA oversight, oncology and patient safety, and AI and emerging technology [41, 42].

Committee referral. Expect primary referral to the Committee on Energy and Commerce (Subcommittee on Health), which holds jurisdiction over the Federal Food, Drug, and Cosmetic Act and FDA device authorities. Because the bill addresses machine-generated robot-patient interaction code and the autonomy of Physical AI systems, a secondary referral to the Committee on Science, Space, and Technology is possible. The Speaker, advised by the Parliamentarian, makes the referral after introduction; the sponsor's staff may ask the Parliamentarian to review the draft in advance for likely referral and parliamentary problems.

Introduction checklist. Each item is confirmed before the bill is filed.

- ☐ One authoritative introduction version of the text is selected, frozen, and version-controlled, with no further edits to the introduced version.
- ☐ A named human attorney and qualified oncology and regulatory experts have reviewed and taken ownership of the final text.
- ☐ Currency check completed: no newer public law has altered Title 21, the target section number (515D, 21 U.S.C. 360e-5) remains unused, and every cross-reference and conforming amendment still lands.
- ☐ Final text routed through the House Office of the Legislative Counsel for proper legislative form.
- ☐ Parliamentarian review requested, through the sponsor's staff, for likely referral and parliamentary issues.
- ☐ Sponsor identified and committed, and the sponsor's signature obtained on the bill.
- ☐ Constitutional Authority Statement finalized, signed, and ready to submit with the measure (Article I, Section 8, Clause 3, supported by Clause 18).
- ☐ Original cosponsor list collected on the exact introduction version, with bipartisan balance pursued.
- ☐ Companion materials assembled: one-pager, section-by-section, policy memo, findings, and Ramseyer print.
- ☐ PAYGO statement and earmark declaration prepared as applicable; the bill directs rulemaking only and is expected to be PAYGO neutral.
- ☐ Filing logistics confirmed: submission via the eHopper, the House's primary electronic introduction mechanism, accepted from about 15 minutes before convening through about 15 minutes after adjournment.

□ AI-assisted origin disclosed to the sponsoring office, with a named human owner taking responsibility for the final text.

Sign-on. The first signer is the sponsor (one only). All subsequent signers committing before or at introduction are original cosponsors. These lines are illustrative placeholders; no Member has agreed to sign, and only a sitting Member may introduce the measure.

Sponsor (Representative), date

Original cosponsor (Representative), date

Original cosponsor (Representative), date

Transparency note. This draft and its companion materials were AI-assisted, generated with Claude Code (Opus 4.8). No House rule requires disclosure that text was AI-drafted, but the sponsoring office should be told plainly, and a named human attorney and subject-matter experts should review the text and take ownership of it before introduction. Until a Member actually introduces the bill, it remains an independent research draft: not pending legislation, not endorsed by the FDA, HHS, or any Member of Congress, and not legal advice [46].

APPENDIX I. STAKEHOLDER ENGAGEMENT PLAN (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, any standards body, or any Member of Congress. Bill: H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session. Refer to: House Committee on Energy and Commerce (Subcommittee on Health). Scope: the pre-introduction engagement step that precedes any sponsor's introduction.

Goals. The operative rule of H. R. 9510 is that no robot-patient interaction code may be generated or executed in a Physical AI oncology investigation unless an automated verification, validation, and uncertainty quantification process, bound to named external consensus standards, has first cleared. Because the gate is bound to specific published standards, the bodies that author and maintain those standards are the natural technical reviewers of whether the bindings are stated correctly. Outreach therefore has four goals: test the standards bindings for accuracy and currency with the people who wrote them; surface implementation burden and feasibility concerns early, while the text can still change; build informed, voluntary support among the communities the bill affects; and keep a faithful, traceable record of who was consulted and what they said.

Principles. Five principles govern every contact. First, accuracy before advocacy: lead with the text and the evidence, not with persuasion. Second, no claimed endorsement: no group has endorsed this bill, and no message, slide, or email may imply otherwise. Third, listen to revise: treat objections as inputs to the draft, not obstacles to it. Fourth, transparency of origin: the draft was AI-assisted, and that is disclosed up front so a named human owner can take responsibility for the final text. Fifth, respect for role: standards bodies set consensus standards and do not legislate, agencies administer the law and do not author bills, and patient groups speak for patients; the plan never asks any party to act outside its role [46].

Sequencing. Engagement runs as a sequenced wave inside calendar year 2026, because the 119th Congress, 2d Session ends January 3, 2027, and a draft not introduced and acted on by then must

be reintroduced. Preparation comes first: finalize the briefing package, confirm each standards citation against the most current edition, and stand up the tracking log. Technical review with the standards bodies follows, with ASME approached first because ASME V&V 40-2018 and the in-development VVUQ 70 are the keystone bindings, then the medical-device and robot-safety bodies, then the remaining standards organizations; these contacts are corrective in purpose, confirming the bindings before anyone is asked to support the text. Agency and community briefings come next, in parallel: FDA (CDRH) and HHS on administrability, and patient-advocacy, oncology, industry, and academic groups on the substance. Consolidation folds feedback into a single revised text and reconfirms that the bindings still resolve. Handoff provides the consolidated text and the engagement record to a prospective House sponsor's office, with all disclaimers intact. Standards-body review precedes the support-building briefings on purpose: the bill stands or falls on whether its external anchors are stated correctly, so that has to be verified before the text is socialized more broadly [7, 47].

Risks and respect. The principal risk is implied endorsement: the bill names many organizations, and any of them could reasonably worry that being briefed looks like backing the bill. Mitigation is that every contact opens and closes with the standing disclaimer, the deck carries it on the title slide, and follow-up email repeats that no endorsement is sought or implied. A second risk is misstating a standard, which would be both inaccurate and disrespectful to the body that authored it; mitigation is the corrective framing above, where standards bodies are asked to correct the bindings before the text is advocated, plus a final pre-handoff citation check. A third risk is overstepping role, for example asking an agency to take a position on pending-style legislation it should not comment on; mitigation is to confine each ask to what fits that party's role. A fourth risk is opacity about origin; mitigation is to disclose the AI-assisted drafting and the named human owner in the first contact. A final risk is fatigue from repeated outreach; mitigation is a single consolidated package, one primary contact per organization, and no second approach unless invited. Throughout, disagreement is recorded faithfully and reflected in the text rather than argued away.

Table 8 is the stakeholder register: for each stakeholder it records the principal interest, the message carried to that audience, and the engagement phase in which the contact occurs. The message column is consistent with the not-endorsed-by framing; the request to a standards body is technical accuracy, not support.

Stakeholder	Interest	Message	Phase
FDA (CDRH) and HHS	Administrability of a new §515D (21 U.S.C. 360e-5) and its fit with premarket review and predetermined change control plans (§515C)	A sequencing rule that complements, not duplicates, the existing PCCP and quality-system framework; the ask is the administrability read a new §515D would require	Agency
Patient-advocacy and oncology groups	Patient safety, informed consent, and equitable access in trials with embodied AI	A patient-protective rule that requires verification before an irreversible motion; where should disclosure and protection be stronger	Community
Medical-device and AI industry	Feasibility, cost, timelines, and interoperability with existing pipelines	A predictable, standards-anchored rule with a realistic transition window; help confirm the transition periods are workable	Community
Academic and professional societies	Scientific rigor, reporting completeness, and evidence quality	A rule grounded in the consensus methods your members publish; help keep the methodology current	Community

Stakeholder	Interest	Message	Phase
ASME (V&V 40, VVUQ 70)	The keystone bindings: ASME V&V 40-2018 credibility commensurate with risk, and VVUQ 70 machine-learning V&V in development	The gate operationalizes the credibility-commensurate-with-risk logic your committees defined; please confirm the references and VVUQ 70's development status	Technical
IEC (80601-2-77, 60601-1, 62304)	Medical-device and software safety: essential performance, single-fault, and software life cycle	The gate protects the essential performance your standards already define for surgical robots; confirm the editions and amendments match	Technical
ISO (13482, ISO/TS 15066, 14971, 13849-1, 9283)	Robot safety and metrics: human-contact force limits, risk management, control safety, and pose accuracy	The gate verifies the measurable thresholds your standards specify for robot-patient contact; confirm the bindings are correctly scoped	Technical
IEEE (7009-2024)	Fail-safe autonomy and hand-back-to-human escalation	The bill's escalation logic mirrors your fail-safe design standard; confirm the escalation rule is consistent with 7009-2024	Technical
NASA (NASA-STD-7009A)	Uncertainty quantification as a first-class credibility factor	UQ is a named, first-class gate criterion, consistent with your models-and-simulations standard; confirm UQ is treated as your standard intends	Technical
UL (UL 4600)	Safety case for autonomous products	The gate produces and checks the structured claim-argument-evidence safety case your standard frames; confirm the artifact aligns with UL 4600	Technical
AMA (CPT Appendix S)	Coding and the assistive, augmentative, autonomous AI taxonomy	The bill's autonomy vocabulary is consistent with your AI taxonomy; confirm the framing aligns with Appendix S	Technical

Table 8. Stakeholder register for H. R. 9510: the principal interest, the message carried, and the engagement phase for each audience. Standards-body contacts are corrective and seek technical accuracy, not endorsement; no group has endorsed this measure [48, 10, 39, 49, 50, 9, 51, 20].

APPENDIX J. LEGISLATIVE COUNSEL ROUTING MEMO (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, the House Office of the Legislative Counsel, or any Member of Congress. This memorandum is non-operative; it transmits a draft for legislative-form review and carries no legal force of its own.

TO: House Office of the Legislative Counsel, via the sponsoring office (Member of the House Committee on Energy and Commerce, Subcommittee on Health)

FROM: CEO Kevin Kawchak, ChemicalQDevice

RE: Drafting review and routing of H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session, the Verification Before Generation in Physical AI Oncology Trials Act of 2026, amending the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 301 et seq.), current through Public Law 119-93

DATE: June 7, 2026

Purpose and routing. This memo transmits a research draft of H. R. 9510 for legislative-form review and asks the House Office of the Legislative Counsel (HOLC) to confirm or correct the draft's amendatory structure before introduction. HOLC provides nonpartisan, impartial, and confidential drafting services to House Members and committees, not directly to the public, so this proposal is submitted through the sponsoring Member's office, the proper channel for engaging HOLC. We expect that several drafts may be required, and we ask that HOLC treat the attached text as a starting draft to be revised through the office, not a finished product [45, 52].

The draft is in bill form and can become law after passage by both chambers and presentment to the President; it is not a simple or concurrent resolution. The designation H. R. 9510 is a placeholder; the Clerk assigns the operative number only at introduction. The bill is paired with companion deliverables in this package, including a section-by-section analysis, a Ramseyer-style comparative print, legislative findings, a constitutional authority statement under Rule XII, clause 7(c)(1), and a plain-English policy memo. Those documents are supporting matter, not part of the bill text submitted for codification.

Vehicle and positive-law question. The bill amends the Federal Food, Drug, and Cosmetic Act (FD&C Act), 21 U.S.C. 301 et seq., current through Public Law 119-93. The central form question is positive-law status. The FD&C Act has not been enacted into positive law as Title 21; Title 21 is a compilation of which the FD&C Act is the underlying and legally controlling text. The draft therefore amends the underlying Act by the Act’s own section numbers (for example, “Section 515D of the Federal Food, Drug, and Cosmetic Act is amended”), giving the corresponding Code citation parenthetically (for example, “(21 U.S.C. 360e-5)”). HOLC drafting guidance notes that positive-law status governs whether a measure amends the Code provision or the underlying statute. We ask HOLC to confirm that this Act-amends-Act approach, with parenthetical 21 U.S.C. citations, is correct and consistently applied [45, 53].

Amendatory structure. Section 3 of the bill is the operative provision. Subsection (a) inserts a new section 515D into subchapter V of the Act, immediately after section 515C; subsection (b) makes the ten conforming amendments; subsection (c) is the clerical amendment to the Act’s table of contents; subsection (d) is an Act-level rule of construction and savings provision; and subsection (e) carries the effective date, the 365-day rulemaking deadline, and the 180-day transition rule. Table 9 lists the new section and each conforming amendment, with its FD&C Act section, the parenthetical 21 U.S.C. citation, and the amendatory instruction [32, 38].

Provision	FD&C	21 U.S.C.	Amendatory instruction
New operative section	§ 515D	§ 360e-5	Insert new section 515D in subchapter V after section 515C, with subsections setting the verification, validation, and uncertainty-quantification gate requirements
Conforming (1)	§ 201(h)	§ 321(h)	Confirm that robot-patient interaction code and autonomous robot-control software are a device
Conforming (2)	§ 301	§ 331	Add a prohibited act for generation or execution in violation of section 515D and for failure to document and attest
Conforming (3)	§ 501	§ 351	Deem a Physical AI device software function used without a cleared section 515D record to be adulterated
Conforming (4)	§ 505F	§ 355g	Recognize section 515D records as real world evidence for the program
Conforming (5)	§ 510(k)	§ 360(k)	Provide that a verification-governed change under an established PCCP needs no new premarket notification
Conforming (6)	§ 513	§ 360c	Graduate special controls by the device’s degree of autonomy
Conforming (7)	§ 515	§ 360e	Require a PMA application to include the cleared verification record and gate evidence
Conforming (8)	§ 515C	§ 360e-4	Require a predetermined change control plan to include an automated verification-before-change protocol
Conforming (9)	§ 520(o)	§ 360j(o)	Confirm such software is not within the clinical-decision-support exclusion
Conforming (10)	§ 521	§ 360k	Add a savings clause preserving State human-in-the-loop and audit requirements
Clerical	Contents	(Act table)	Insert the “Sec. 515D” item after the item for section 515C

Table 9. The amendatory-instruction table: the new operative section, the ten conforming amendments, and the clerical amendment, each with its FD&C Act section, its 21 U.S.C. citation, and the instruction.

We ask HOLC to confirm the amendatory verbs, the quotation form, and the placement of inserted matter, in particular the “is amended by inserting after section 515C the following:” construction and the quoted block for new section 515D, and that each conforming amendment uses the correct point of insertion within its existing subsection rather than a freestanding restatement.

Cross-reference and disambiguation. The bill turns on a section-number distinction that is easy to misread, and we flag it for special attention. Section 360(k) of Title 21 is the premarket notification (510(k)) pathway, referenced in conforming amendment (5) and inside new section 515D and section 515C. Section 360k of Title 21, with no parenthetical, is a different and freestanding provision, State and local requirements respecting devices (preemption), amended by conforming amendment (10) to add the savings clause. The two are not the same section and must not be conflated. We have used “section 360(k)” only for the 510(k) pathway and “section 360k” only for the preemption provision throughout the draft and the comparative print. We ask HOLC to verify that every internal cross-reference, including references from new section 515D to sections 515C, 515, 513, and 520(g), and to the part 11, part 50, and part 312 frameworks, points to the intended target, and that the conforming “; and” edits at the points of insertion match the current text.

Definitions and effective dates. New section 515D defines the operative terms, among them Physical AI system, robot agent, robot-patient interaction code, generative artificial intelligence, AI-enabled device software function, verification, validation, uncertainty quantification, verification fraction, validation agreement, the coefficient-of-variation bound, the hard catastrophe predicate, Physical AI Standard Level, Unification Standard Level, and hash-chained audit trail. We ask HOLC to advise on reconciling these section-specific definitions with the Act’s general device definition in section 201(h) as amended, and the new software-function terms with the clinical-decision-support exclusion in section 520(o) as amended, so the local definitions neither contradict nor are silently overridden by the general ones.

On timing, section 3(e) provides that the Act takes effect on the first day of the first calendar quarter beginning not less than one year after enactment; that the Secretary must issue final implementing regulations not later than 365 days after enactment; and that an investigation already enrolling on the effective date has 180 days to come into compliance. The provision leaves the February 2, 2026, Quality Management System Regulation effective date undisturbed. We ask HOLC to confirm that the effective date, rulemaking deadline, and transition rule are mutually consistent and free of gaps [6].

The pre-introduction path. Figure 6 traces the route by actor. The sponsoring office consolidates one authoritative text; experts and counsel conduct human and legal review; the currency check in Appendix K re-pulls every anchor; HOLC puts the measure in legislative form and attaches the constitutional authority statement and the PAYGO note; the office recruits a sponsor and original cosponsors; and the measure is introduced through the eHopper, after which the Clerk assigns the number and refers it to Energy and Commerce.

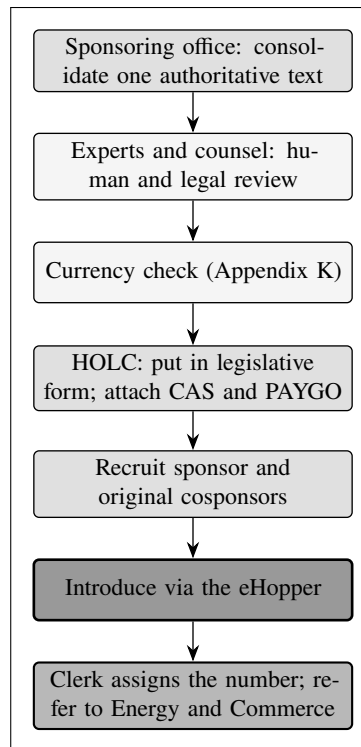


Figure 6. The pre-introduction path by actor: consolidate, review, currency-check, route through the Office of the Legislative Counsel, recruit support, and introduce.

Specific questions for Legislative Counsel. We ask HOLC to address the following form questions before the draft is suitable for introduction.

- (1) Should the bill amend the underlying FD&C Act by its own section numbers with parenthetical 21 U.S.C. citations, given that the Act is not positive-law Title 21, and is that convention applied consistently throughout the draft?
- (2) Are the amendatory instructions and quotation form for new section 515D and for each of the ten conforming amendments in correct legislative form?
- (3) Is the clerical amendment correctly framed to insert the “Sec. 515D” item after the section 515C item in the Act’s table of contents?
- (4) Are the section-number usages correct and unambiguous throughout, in particular section 360(k) (the 510(k) premarket notification pathway) versus section 360k (State and local preemption)?
- (5) Do the section 515D definitions need express reconciliation with section 201(h) and with section 520(o), and if so, what coordinating language do you recommend?
- (6) Are the effective date, the 365-day rulemaking deadline, and the 180-day transition rule internally consistent and properly drafted?
- (7) Should the savings clause in conforming amendment (10) and the rule of construction in section 3(d) be coordinated to avoid any conflict at the preemption boundary under section 360k?
- (8) Does the rule of construction correctly preserve the section 520(g) investigational device exemption, the part 50 and part 312 frameworks, and the section 515C authority, without inadvertently narrowing them?

- (9) Are there any drafting conventions, defined-term placements, or cross-reference forms HOLC would change before this draft is suitable for introduction?

Next steps. We request HOLC’s drafting review through the sponsoring office and will incorporate any revisions into a new draft, preserving a redline, version history, and a short explanation of any unresolved policy choices. After HOLC’s form review, the sponsoring office intends to seek procedural and jurisdictional review from the House Parliamentarian’s office on likely committee referral and any points of order, confirm the sponsor and original cosponsors, finalize the Rule XII constitutional authority statement, and proceed through the current eHopper workflow at introduction. Only the final text approved by the sponsor’s office will be filed [42, 54].

APPENDIX K. CURRENCY AND CROSS-REFERENCE MATRIX (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, the Office of the Law Revision Counsel, or any Member of Congress. This appendix is a verification record, not an assertion that the law cannot change.

Scope. This matrix records the currency and cross-reference audit for H. R. 9510 (illustrative number; the Clerk assigns the real number at introduction), 119th Congress, 2d Session, the Verification Before Generation in Physical AI Oncology Trials Act of 2026. The bill amends the Federal Food, Drug, and Cosmetic Act (FD&C Act), 21 U.S.C. 301 et seq., as reproduced from Title 21, United States Code, current through Public Law 119-93 as of the draft date, June 7, 2026. The operative mechanism is a new section 515D of that Act, codified at 21 U.S.C. 360e-5, inserted immediately after section 515C (21 U.S.C. 360e-4) [38, 53, 32].

Purpose and method. The draft was frozen and currency-checked through Public Law 119-93, and the legal crosswalk that anchors its citations is current through May 31, 2026. This matrix performs the currency task that the next-steps memo identifies as a precondition to introduction: re-verify, immediately before the bill is introduced, that no newer public law has changed Title 21 since Public Law 119-93; that the target section number 515D, codified at 21 U.S.C. 360e-5, is still unused; and that every cross-reference and each of the ten conforming amendments still lands on existing, correctly numbered law. The method is documentary and reproducible. Each row names the authority it depends on, the status confirmed as of June 7, 2026, and the precise action to repeat at introduction. The controlling currency sources are the Office of the Law Revision Counsel (OLRC) United States Code, the text of Public Law 119-93, and the legal crosswalk current through May 31, 2026 [53, 55].

Title 21 is amended frequently, so the value of the matrix is that it tells a sponsoring office exactly which anchors to re-pull, and what a clean result looks like, on the day the measure is filed. Table 10 is the verification matrix: each row gives the citation, what to verify, and its status as of the draft date.

Citation	What to verify	Status (June 7, 2026)
Title 21, chapter 9	No public law after Public Law 119-93 has amended the FD&C Act chapter (21 U.S.C. 301 et seq.)	Current through Public Law 119-93; no later amendment reflected
§ 301 (note)	The new Short Title of 2026 Amendment note is additive and does not collide with the existing 2025 note	2025 note present; 2026 note open

Citation	What to verify	Status (June 7, 2026)
§ 331	Last lettered prohibited act is (iii), so the new act lands as (jjj)	Last act confirmed at (iii); (jjj) open
§ 351	Subsections run through (j), so the new act lands as (k)	(h), (i), (j) present; (k) open
§ 355g	Subsections (a) and (b), so the new act lands as (g)	(a), (b) confirmed; (g) open
§ 360e(c)(1)	Clauses run through (H), so the new clause lands as (I)	(G), (H) confirmed; (I) open
§ 360c	Subsections run through (k), so the new act lands as (l)	Through (k) confirmed; (l) open
§ 360e-4	Subsections run through (c), so the new act lands as (d)	Through (c) confirmed; (d) open
§ 360k	Subsections (a) and (b), so the savings clause lands as (c)	(a), (b) confirmed; (c) open
§ 360e-5 (§ 515D)	The target number is unassigned in both the Code and the FD&C Act table of contents	Confirmed unused; well placed after section 515C
Legal crosswalk	The crosswalk anchoring the citations has not been revised in a way that moves an anchor	Current through May 31, 2026

Table 10. The currency and cross-reference verification matrix: each citation, what to verify, and its status as of the draft date. A clean result at introduction is every row confirmed and section 360e-5 still unused.

Target-number availability. The Act needs one vacant section number coordinated across two systems: the FD&C Act's own section numbering and the United States Code classification. The chosen pair is FD&C Act section 515D, classified to 21 U.S.C. 360e-5. The premarket-approval cluster runs section 515 (21 U.S.C. 360e), section 515A (21 U.S.C. 360e-1), section 515B (21 U.S.C. 360e-2), and section 515C (21 U.S.C. 360e-4), with the predetermined change control plan authority at section 515C. Inserting section 515D, codified at 21 U.S.C. 360e-5, immediately after section 515C places the new verification authority next to the change-control authority it most closely resembles, and the suffix is the next open slot in the section 360e family. As of June 7, 2026, neither section 515D nor 21 U.S.C. 360e-5 is assigned in Title 21. To re-verify at introduction, search the OLRC United States Code for 360e-5 and the FD&C Act table of contents for 515D; both must return no existing section. If either has been taken by an intervening law, renumber to the next open suffix and re-run the conforming-amendment matrix, because every conforming amendment cross-references section 515D [32, 53].

Section-number disambiguation. A recurring drafting hazard in Title 21 device law is the near-identical pair section 360(k) and section 360k. They are different provisions and must not be conflated. Section 360(k) is subsection (k) of section 360 (FD&C Act section 510(k)), the premarket-notification authority, on which the section 510(k) conforming amendment and the change-control cross-references in section 515C rely. Section 360k is a separate, freestanding section (FD&C Act section 521) on State and local requirements respecting devices, that is, device preemption, to which the Act adds a new savings clause. Every operative use of 510(k) in the bill cites 21 U.S.C. 360(k) with the parenthetical, and every use of the preemption section cites 21 U.S.C. 360k without parentheses [29, 34].

Build lineage. The currency record sits within a larger build history. Figure 9 places the prompt evolution beside the bill-version evolution: a sequence of prompts (next-steps, figures-bill, draft-bill,

full-bill, final-bill) runs alongside the bill versions (v1.0 from VVUQ-03, v2.0 from VVUQ-04, v3.0 from VVUQ-05, and v4.0, this build), and the final-bill prompt produces this v4.0 build, the fourth bill perspective [55, 56].

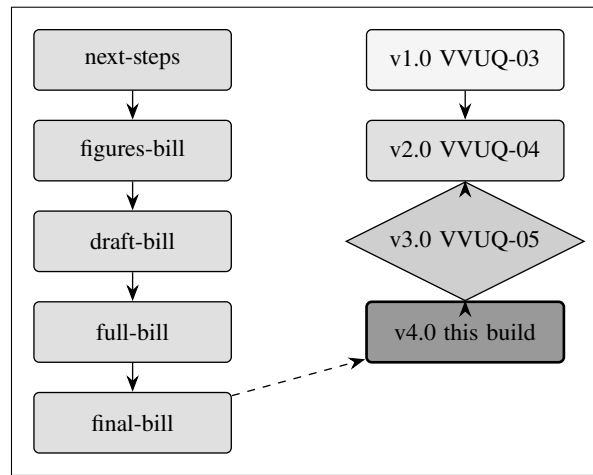


Figure 9. Prompt evolution beside bill-version evolution: the final-bill prompt produces this v4.0 build, the fourth bill perspective.

Re-verification checklist before introduction. Each item is a clean result the verifier of record confirms on the day the measure is filed.

- (1) Pull the OLRC United States Code currency banner for Title 21 and confirm the latest public law affecting chapter 9 is not newer than Public Law 119-93.
- (2) Read the full text of Public Law 119-93 and confirm it does not itself disturb any of the ten conforming anchors.
- (3) Confirm 21 U.S.C. 360e-5 (FD&C Act section 515D) is still unassigned in both the Code and the FD&C Act table of contents.
- (4) Re-read 21 U.S.C. 331 and confirm the last lettered prohibited act is still (iii), so (jjj) remains open.
- (5) Confirm 21 U.S.C. 351 still runs through (j); section 355g through (b); section 360e(c)(1) through (H); section 360c through (k); section 360e-4 through (c); section 360k through (b).
- (6) Confirm the device-definition exclusion language in 21 U.S.C. 321(h) and the clinical-decision-support exclusion at 21 U.S.C. 360j(o)(1)(A) through (E) are unchanged.
- (7) Confirm every 510(k) cite reads 21 U.S.C. 360(k) and every preemption cite reads 21 U.S.C. 360k.
- (8) Confirm the conforming inserted-punctuation edits at 21 U.S.C. 360e(c)(1)(G) and (H) still match the current text.
- (9) Record the verification date, the OLRC currency string, and the verifier of record; attach the result to the frozen bill text before the sponsoring office files via the eHopper.

Sign-off. The audit is complete only when a named human verifier signs the result against the frozen introduction text and confirms, as of the filing date, that the draft remains current through Public Law 119-93 and that section 515D, codified at 21 U.S.C. 360e-5, is still unused. These lines are illustrative placeholders.

Verifier of record, with OLRC currency string, date

Sponsoring office staff confirmation, date

APPENDIX L. TESTIMONY AND RESEARCH-INFLUENCE BRIEF (NON-OPERATIVE).

Independent research draft. Not an enacted law, not pending legislation, and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress. The bills and executive actions discussed are research influences only and are not the legal basis of any operative provision.

This appendix has two parts. Part I is hearing testimony, written to be read aloud in about five minutes before the House Committee on Energy and Commerce (Subcommittee on Health), with a question-and-answer block. Part II is a research-influence brief that records, for the hearing record, exactly how recent executive actions and emerging 119th Congress bills relate to H. R. 9510, and exactly why none of them appears in its operative text.

PART I. HEARING TESTIMONY.

Chairman, Ranking Member, and Members of the Subcommittee: thank you for the opportunity to testify on the verification of artificial intelligence in cancer trials, and specifically on the safety of AI that does not merely advise a clinician but physically moves around a patient and acts upon that patient.

Who I am. I am Kevin Kawchak, Chief Executive Officer of ChemicalQDevice. Our work concerns the verification, validation, and uncertainty quantification, or VVUQ, of high-consequence AI systems, including the control software for surgical and patient-contacting robots used in oncology research. I appear in an independent research capacity. The draft bill I will describe, H. R. 9510, is an illustrative research instrument; no Member has agreed to sponsor it, and nothing I say should be taken as the position of the FDA, of HHS, or of any Member of Congress.

The problem. Cancer trials are beginning to place robots in direct physical contact with patients: surgical humanoids that cut and suture, positioning systems that hold a patient during treatment, and platforms that place biopsy or delivery needles. What such a robot does at the instant it touches a patient is decided by software, and that software is increasingly written by AI in real time rather than typed, line by line, by a human engineer. Here is the gap. No Federal law governs the order of operations. A trial sponsor today may generate the robot-patient interaction code, run it on a patient, and document the safety checks afterward, in whatever order it chooses, because the order is simply not regulated. We call this the sequencing gap. Sequence matters because of embodiment. In an ordinary data pipeline, a software error produces a wrong number that a human can catch downstream before anyone is harmed. In an embodied surgical system, the same class of error produces a wrong physical motion that a human cannot catch in time, because one moving body concentrates every failure mode into a single instrument acting in real time. Checking the code after it has already moved is, in this setting, checking too late.

The proposal. H. R. 9510 supplies the one missing rule. It amends the Federal Food, Drug, and Cosmetic Act and adds a single new section, 515D, codified at 21 U.S.C. 360e-5, placed immediately after the predetermined change control authority at section 515C that it most resembles. The core rule is one sentence of policy: verify before generate. No robot-patient interaction code may be generated or executed in a Physical AI oncology clinical investigation unless an automated VVUQ process, bound to named external standards, has first cleared that interaction, with the record documented and attested. A record created after generation or execution does not satisfy the section.

The timing of the record is itself the compliance fact. The gate is concrete, not aspirational. Each check returns one of three results: ACCEPT, BLOCK, or ESCALATE, which returns the interaction to a qualified human. For each interaction the verification fraction must equal 1.0, meaning every check passes, with no partial credit. Three gates carry a hard catastrophe predicate at perfect agreement: no breach of a vascular no-fly volume, a guaranteed minimum clearance from any human sharing the room, and a guaranteed safe state on fault or emergency stop [32].

Why it is feasible. This is not a hope; it has been run. In a first study, VVUQ-01, an automated pipeline held the verification fraction at 1.0 across 51 of 51 tests, with 1 ACCEPT and 5 BLOCK outcomes. In a second study, VVUQ-02, a ten-gate assurance suite over the control software for a surgical humanoid cleared 172 of 172 tests, with a composite mean of 93.56, reproducibly from a fixed seed, 20260525. The bill asks sponsors to do, as a precondition, what has already been demonstrated to work. The gate is bound to recognized standards rather than to anyone's discretion: its thresholds tie to ASME V&V 40, IEC 80601-2-77, ISO/TS 15066, ISO 14971, NASA-STD-7009A, and IEEE 7009. That binding is what makes the test auditable and what keeps it from becoming a moving target [3, 4].

Why it fits existing law. I want to be candid about the policy environment. The current Federal posture toward AI is deliberately deregulatory and pro-innovation. This bill is designed to complement that posture, not to fight it. It is not a broad AI mandate. It is a narrow, standards-bound safety gate for one thing: embodied robot-patient code in clinical trials. The bill is additive. It leaves the investigational-device exemption, human-subject protection, and the investigational new drug framework in place. It frames the gate as a pre-authorized change-control discipline of the kind section 515C already contemplates, routes its records through the electronic-records rule and the quality-system regulation that sponsors already meet, and adds a savings clause so existing State human-in-the-loop laws are preserved. It creates no new agency and no new spending program; it directs rulemaking only [57, 58].

Respectful asks. I respectfully ask the Subcommittee to consider three things. First, treat the sequencing gap as a discrete, well-defined safety question, separate from the larger and more contested debates about AI policy generally. Second, recognize that verification before generation is technically achievable today, on the evidence I have summarized, so the question before you is one of timing and assurance, not of feasibility. Third, if the Subcommittee acts, keep any measure narrow and standards-bound, so that it strengthens the credibility of American oncology research without becoming the broad mandate that current Federal policy disfavors. Thank you for your attention. I welcome your questions.

Anticipated questions and answers. The following exchange anticipates the questions a Member is most likely to ask, with the responses the witness is prepared to give.

Q. Is this a broad new mandate on artificial intelligence?

A. No. The section reaches one thing only: robot-patient interaction code in a Physical AI oncology clinical investigation. It does not touch disembodied diagnostic software, clinical-decision-support tools that only advise a clinician, or AI outside the trial setting. It is a single sequencing rule, bound to named external standards, for code that physically moves around a patient.

Q. Will this slow down American innovation in cancer research?

A. It is designed not to. The gate is automated, so it runs at machine speed rather than waiting on manual review, and it asks sponsors to do as a precondition what two studies already show is achievable. By making the verification record auditable and standards-bound, it strengthens the credibility of the research it governs, which is what allows results to be trusted and relied upon.

Q. Why legislate now rather than leave this to FDA guidance?

A. The closest existing analogue, the predetermined change control plan authority in section 515C and its final guidance, is voluntary, submission-based, and neither automated nor required in advance of generation. Guidance can describe good practice, but it cannot establish the order-of-operations rule that embodiment makes necessary. The bill supplies that one rule and directs the Secretary to issue final implementing regulations within 365 days [32].

Q. Does the bill conflict with the current deregulatory posture or with State law?

A. It is built to complement the deregulatory posture: it is narrow, standards-bound, and adds no new agency or spending program. On State law, it includes a savings clause that preserves existing State human-in-the-loop and audit requirements, so it sets a Federal floor for one safety question without displacing the human-oversight protections States have already enacted [59].

PART II. RESEARCH-INFLUENCE BRIEF.

This brief records, for the hearing record, how each recent executive action and each emerging 119th Congress bill relates to H. R. 9510, and demonstrates that none of them is, or could be, the legal basis of any operative provision. Table 11 is the research-influence matrix; the placement column states where each authority is used, and the operative-versus-influence rule keeps every one of them out of the statutory text and the comparative print.

Authority	What it is	What it informs	Placement
EO 14179	Removing Barriers to American Leadership in AI (90 FR 8741)	Deregulatory posture in the findings narrative; the gate framed as credibility rather than mandate	Memo; matrix; testimony
EO 14365	Ensuring a National Policy Framework for AI	State-law review and preemption context; the case for one national documentation standard	Appendix; matrix; testimony
OMB M-25-21	Accelerating Federal Use of AI (pre-deployment testing)	The pre-deployment-testing analogue; the Federal lifecycle the gate maps onto	Memo; matrix; testimony
HHS AI Strategy	The HHS Artificial Intelligence Strategy (Dec. 2025)	Governance and reproducibility framing that resonates with VVUQ goals	Memo; matrix
H.R. 238	Healthy Technology Act of 2025	AI prescribing-authority direction; why autonomous clinical AI raises the evidentiary bar	Testimony; matrix
S. 1399	Health Tech Investment Act	Medicare payment-class direction; clearance-gated payment makes validation the entry ticket	Testimony; matrix
H.R. 6361	Ban AI Denials in Medicare Act	The human-over-AI signal; bipartisan distrust of opaque AI decisions	Findings narrative; matrix
H.R. 5045	HEALTH AI Act	Generative-AI research direction; potential funding for validation methodology	Memo; matrix
H.Res. 694	Sense of the House on the CMS WISer AI pilot	The countervailing human-review signal; demand for auditable AI	Findings narrative; matrix

Table 11. The research-influence matrix: emerging 119th Congress bills and executive actions, what each informs, and where each is used. The placement column is memo, appendix, or testimony only; none is ever in the operative text.

How each influence relates, and how it stays out of operative text. The executive actions set the environment the bill is drafted into. EO 14179 and EO 14365 together describe a deregulatory, preemption-minded Federal posture; they influence the bill in one direction only, arguing against a broad mandate and in favor of a narrow, voluntary-credibility framing and a single national documentation standard. OMB M-25-21 is more structural: its requirement of pre-deployment testing, an impact assessment, and ongoing monitoring is the cleanest surviving Federal articulation of a VVUQ-style lifecycle, used as an analogue to show that verifying before deployment is already a recognized Federal idea. The HHS Artificial Intelligence Strategy of December 2025 is invoked

only to show alignment of values. None of these is operative authority, and the bill never treats them as such. They are executive and sub-regulatory instruments, not statutes; they can change with an administration; and a safety gate that drew its legal force from an executive order would be unstable by construction. The bill therefore takes its operative force from the Federal Food, Drug, and Cosmetic Act and from recognized consensus standards [57, 59, 58, 60].

The emerging bills establish legislative momentum on both sides of one question. On the autonomy side, H.R. 238 would let AI qualify as a practitioner eligible to prescribe, and S. 1399 would create a Medicare new-technology payment class for cleared algorithm-based services; both raise the stakes of getting autonomous clinical AI right, which is the gap a verification gate addresses. On the human-oversight side, H.R. 6361 would bar AI-driven prior authorization in traditional Medicare, and H.Res. 694 urges CMS to halt an AI coverage pilot; both reflect a bipartisan insistence that a human stay accountable for consequential AI decisions. H.R. 5045 sits alongside these as a research-funding vehicle that could support validation and evaluation methodology. These bills inform the testimony, the findings, and the matrix, and nothing more. They are introduced or in-committee measures; not one is enacted, and an introduced bill has no legal effect, so citing one as the basis of an operative clause would be both incorrect and misleading [61, 62, 63, 64, 65].

Verification note. Each authority above was verified by number, subject, and status before use, and each is presented strictly as a research influence; readers should re-verify currency before relying on any entry. Three cautions govern this brief. First, there is no valid Act in the 119th Congress on this subject; the measures titled VALID Act of 2025 are an unrelated matter, so no such Act is cited here. Second, a standalone AI-in-clinical-trials bill and a standalone Federal health-AI liability bill could not be verified, and no bill numbers are assigned to them. Third, the sponsor of H.Res. 694 was not cleanly confirmable, so it is cited by number and subject only. Consistent with the whole package, only enacted statutes, in-force rules, and recognized standards carry operative weight; the bills and executive actions in this brief carry none.

Evidence to introduction. Figure 7 traces the handoff this appendix sits within: the author sets requirements; Claude Code Opus 4.8 generates and runs the work; the VVUQ gates and tests confirm it; the Office of the Legislative Counsel puts the result in legislative form; and a sponsoring Member introduces it through the House eHopper [46, 42].

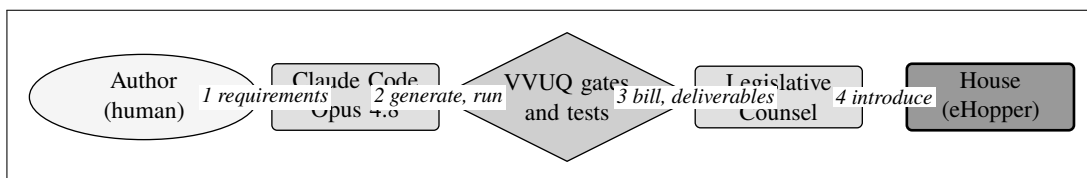


Figure 7. Evidence to introduction handoff: requirements to autonomous generation and gating, to legislative form, to introduction by a sponsoring Member.

AUTHORSHIP, NOTICES, AND PROVENANCE

Prepared by CEO Kevin Kawchak (ORCID 0009-0007-5457-8667), ChemicalQDevice (kevink@chemicalqdevice.com). Adapted using Claude Code Opus 4.8 (1M context) Max; research assisted by ChatGPT and Google Gemini.

Notice: This measure was adapted from documents of the United States Government in the public domain under 17 U.S.C. 105. The reproduced statutory text is in the public domain; the authoritative version is the United States Code as published by the Office of the Law Revision Counsel. This work is not endorsed or sponsored by the FDA, HHS, the OLRC, CFR, ICH, or any member of Congress.

Disclaimer: This work is an independent research draft, is not an enacted law, and is not legal advice; it is not endorsed, sponsored, or approved by any trial sponsor, contract research organization, site, institutional review board, regulator, or medical society. All supporting numbers are simulation or illustrative results from the seed 20260525, and no patient-identifiable information is included.

Rights: The reproduced statutory text is in the public domain; the generated amendment framing, figures, tables, and notices are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0), <https://creativecommons.org/licenses/by/4.0/>.

DOI current bill (v4.0): 10.5281/zenodo.xxxxxxxx

DOI prior bill (v3.0): 10.5281/zenodo.xxxxxxxx

DOI repository: 10.5281/zenodo.xxxxxxxx

San Diego. June 7, 2026.

Provenance and Research Sources

References

- [1] K. Kawchak, “National platform for physical ai oncology trials: Usl and psl readiness standards,” 2026. DOI: 10.5281/zenodo.19244918. The Unification Standard Level and Physical AI Standard Level readiness standards the gates rely on. <https://doi.org/10.5281/zenodo.19244918>.
- [2] K. Kawchak, “Vvuq-05 update-bill figures-bill: verbatim visual catalog of the papers/ corpus,” 2026. The five-file catalog (01 root, 02 VVUQ-01, 03 VVUQ-02, 04 VVUQ-03, 05 VVUQ-04) reproducing every ASCII diagram, timeline, repository tree, and table; the visual source for the figure and table slots. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-05/update-bill/figures-bill>.
- [3] K. Kawchak, “Vvuq-01: Automated verification, validation, and uncertainty quantification of generated code,” 2026. The first study: a complete VVUQ pipeline holding the verification fraction at 1.0; 51 of 51 supporting tests passed. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-01>.
- [4] K. Kawchak, “Vvuq-02: A ten-gate assurance suite for robot-patient interaction code,” 2026. The second study: control software for a surgical humanoid cleared a ten-gate suite bound to fourteen external standards and two clinical baselines, reproducibly from a fixed seed. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-02>.
- [5] U.S. Food and Drug Administration, “Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions,” 2025. Final guidance (Dec. 4, 2024) operationalizing 21 U.S.C. § 360e-4 for AI-enabled device software; the closest existing analogue to the verification-before-change duty. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence>.
- [6] U.S. Food and Drug Administration, “Medical devices; quality system regulation amendments (quality management system regulation), 89 fr 7496,” 2024. Final rule (effective Feb. 2, 2026) revising 21 CFR part 820 to incorporate ISO 13485:2016; the live quality scaffold the adulteration tie leans on. <https://www.govinfo.gov/app/details/FR-2024-02-02/2024-01709>.
- [7] ASME, “Asme v&v 40-2018: Assessing credibility of computational modeling through verification and validation: Application to medical devices,” 2018. FDA-recognized consensus standard; the credibility basis (model risk equals influence times consequence) for the verification, validation, and uncertainty quantification gate. <https://www.asme.org/codes-standards/find-codes-standards/assessing-credibility-of-computational-modeling-through-verification-and-validation-application-to-medical-devices>.
- [8] U.S. Food and Drug Administration, “Assessing the credibility of computational modeling and simulation in medical device submissions,” 2023. Final guidance (Nov. 17, 2023) adopting the ASME V&V 40 credibility argument (model risk equals influence times consequence); the regulatory basis for the gate’s credibility grading. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions>.

- [9] National Aeronautics and Space Administration, “Nasa-std-7009a: Standard for models and simulations,” 2016. Treats uncertainty quantification as a first-class credibility factor (epistemic and aleatory); the basis for the coefficient-of-variation dispersion bounds. <https://standards.nasa.gov/standard/nasa/nasa-std-7009>.
- [10] International Organization for Standardization, “Iso/ts 15066:2016: Robots and robotic devices - collaborative robots,” 2016. Power-and-force-limiting and biomechanical thresholds for robot-patient contact; binds the clearance gates. <https://www.iso.org/standard/62996.html>.
- [11] California State Legislature, “Sb 1120, physicians make decisions act (chapter 879, statutes of 2024),” 2024. Effective Jan. 1, 2025; bars AI in utilization review from replacing a licensed physician’s medical-necessity determination. Cited as a finding of existing human-over-AI practice. https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240SB1120.
- [12] Illinois General Assembly, “Hb 1806, wellness and oversight for psychological resources act (public act 104-0054),” 2025. Effective Aug. 4, 2025; bars AI from independent therapeutic decisions without licensed-professional review. Cited as a finding of existing practice. <https://legiscan.com/IL/bill/HB1806/2025>.
- [13] Texas Legislature, “Hb 149, texas responsible artificial intelligence governance act (traiga),” 2025. Effective Jan. 1, 2026; comprehensive AI governance with a regulatory sandbox and advisory council. A finding of existing practice. <https://legiscan.com/TX/bill/HB149/2025>.
- [14] Maryland General Assembly, “Hb 820, utilization review - use of artificial intelligence (chapter 747),” 2025. Effective Oct. 1, 2025; AI in utilization review must use individual clinical information and may not autonomously deny care. Cited as a finding of existing practice. <https://mgaleg.maryland.gov/mgaweb/legislation/details/hb0820>.
- [15] Centers for Medicare and Medicaid Services, “Medicare advantage and part d final rule; artificial intelligence and algorithm guardrail, 42 cfr § 422.101(c),” 2023. 88 FR 22120 (applicable for coverage beginning Jan. 1, 2024); the strongest federal precedent for mandatory human-over-AI oversight in coverage decisions. <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-422/subpart-C/section-422.101>.
- [16] International Council for Harmonisation, “Ich e6(r3): Good clinical practice,” 2025. The harmonized good-clinical-practice basis for the recorded human review and audit requirements. <https://www.ich.org/page/efficacy-guidelines>.
- [17] Office for Civil Rights, HHS, “Nondiscrimination in the use of patient care decision support tools, 45 cfr § 92.210,” 2024. The in-force anchor for the bias-minimization and accessibility duties (identify-and-mitigate compliance date May 1, 2025). <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-92/subpart-B/section-92.210>.
- [18] United States Congress, “Title vi of the civil rights act of 1964, 42 u.s.c. § 2000d,” 1964. Nondiscrimination basis (Title 42, not Title 26) for the bias-minimization duty. <https://www.law.cornell.edu/uscode/text/42/2000d>.
- [19] United States Congress, “Americans with disabilities act of 1990, 42 u.s.c. § 12101 et seq.,” 1990. Accessibility and accommodation basis (Title 42) for the nondiscrimination duty. <https://www.law.cornell.edu/uscode/text/42/12101>.

- [20] American Medical Association, “Cpt appendix s: Artificial intelligence taxonomy (assistive, augmentative, autonomous),” 2026. The graduated-autonomy spine mapped to special controls; robot-control code sits at the autonomous tier requiring the strictest ex ante gate. <https://www.ama-assn.org/practice-management/cpt/cpt-appendix-s-ai-taxonomy-medical-services-procedures>.
- [21] Office of the Federal Register, “Electronic records; electronic signatures, 21 cfr part 11,” 2026. The record-integrity basis for the hash-chained, tamper-evident audit trail required by section 515D. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11>.
- [22] Office for Civil Rights, HHS, “Hipaa security rule, 45 cfr part 164, subpart c,” 2026. The audit-control (§ 164.312(b)) and integrity (§ 164.312(c)) requirements supporting the audit trail. <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-164>.
- [23] International Electrotechnical Commission, “Iec 60601-1: Medical electrical equipment - part 1: General requirements for basic safety and essential performance,” 2020. The single-fault safe-state requirement underlying the fault and emergency-stop gate. <https://webstore.iec.ch/publication/67497>.
- [24] International Organization for Standardization, “Iso 13849-1:2023: Safety of machinery - safety-related parts of control systems,” 2023. The performance level of the fault and emergency-stop control path. <https://www.iso.org/standard/73481.html>.
- [25] United States Congress, “Definitions; the term “device”, 21 u.s.c. § 321(h) (fd&c act § 201(h)),” 2021. The threshold device definition, redesignated (h)(1) by Pub. L. 116-304; the gate that decides whether robot-patient interaction code is regulated. <https://www.law.cornell.edu/uscode/text/21/321>.
- [26] United States Congress, “Prohibited acts, 21 u.s.c. § 331 (fd&c act § 301),” 2022. The list of prohibited acts to which the Act’s penalties under 21 U.S.C. § 333 attach; the enforcement hook for the new duty. <https://www.law.cornell.edu/uscode/text/21/331>.
- [27] United States Congress, “Adulterated drugs and devices, 21 u.s.c. § 351 (fd&c act § 501),” 2017. The device-quality backbone tying adulteration to current good manufacturing practice and 21 U.S.C. § 360j(f). <https://www.law.cornell.edu/uscode/text/21/351>.
- [28] United States Congress, “Utilizing real world evidence, 21 u.s.c. § 355g (fd&c act § 505f),” 2016. The 21st Century Cures Act real-world-evidence program extended to recognize verification records from Physical AI oncology investigations. <https://www.law.cornell.edu/uscode/text/21/355g>.
- [29] United States Congress, “Registration of producers; premarket notification, 21 u.s.c. § 360(k) (fd&c act § 510(k)),” 2022. The 510(k) premarket-notification pathway at § 360(k); distinct from the State-preemption section. <https://www.law.cornell.edu/uscode/text/21/360>.
- [30] United States Congress, “Classification of devices intended for human use, 21 u.s.c. § 360c (fd&c act § 513),” 2022. The risk-based Class I, II, and III scheme and the substantial-equivalence definition; the home of graduated special controls by autonomy. <https://www.law.cornell.edu/uscode/text/21/360c>.
- [31] United States Congress, “Premarket approval, 21 u.s.c. § 360e (fd&c act § 515),” 2022. The Class III premarket-approval pathway; the application contents amended to include the verification record. <https://www.law.cornell.edu/uscode/text/21/360e>.

- [32] United States Congress, “Predetermined change control plans for devices, 21 u.s.c. § 360e-4 (fd&c act § 515c),” 2022. The keystone change-control authority, added by FDORA, after which the new section 515D (21 U.S.C. § 360e-5) is inserted. <https://www.law.cornell.edu/uscode/text/21/360e-4>.
- [33] United States Congress, “General provisions; the software and clinical decision support exclusion, 21 u.s.c. § 360j(o) (fd&c act § 520(o)),” 2022. The Cures Act software exclusion that robot-control software must not fall through; confirmed not to reach code that acts on a patient. <https://www.law.cornell.edu/uscode/text/21/360j>.
- [34] United States Congress, “State and local requirements respecting devices, 21 u.s.c. § 360k (fd&c act § 521),” 1976. Device preemption; the boundary the savings clause marks for State human-over-AI requirements. Distinct from the 510(k) pathway at § 360(k). <https://www.law.cornell.edu/uscode/text/21/360k>.
- [35] Office of the Federal Register, “Protection of human subjects, 21 cfr part 50,” 2026. Human-subject protection and informed consent the bill preserves unchanged. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-50>.
- [36] Office of the Federal Register, “Investigational new drug application, 21 cfr part 312,” 2026. The IND framework the verification gate attaches to and that the rule of construction preserves. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-312>.
- [37] United States Congress, “Federal food, drug, and cosmetic act, 21 u.s.c. § 301 et seq.,” 1938. The underlying Act this bill amends; the device-regulation backbone reproduced in the comparative print. <https://www.law.cornell.edu/uscode/text/21/chapter-9>.
- [38] United States Congress, “Public law 119-93 (the publication point title 21 is current through),” 2026. The public law through which the reproduced Title 21 text is current; named as the currency point, not as the law being amended. <https://www.govinfo.gov/app/collection/plaw>.
- [39] International Organization for Standardization, “Iso 14971:2019: Medical devices - application of risk management to medical devices,” 2019. The master risk-management process; the gate is a risk-control measure and the hard catastrophe predicate a risk acceptance limit under it. <https://www.iso.org/standard/72704.html>.
- [40] Congressional Budget Office, “Cost estimates and the statutory pay-as-you-go act of 2010,” 2026. The basis for the PAYGO and cost-estimate deliverable; informal CBO consultation can precede a formal score. <https://www.cbo.gov/system/files/2026-06/hr8872.pdf>.
- [41] U.S. House of Representatives, “Rules of the house of representatives, 119th congress (rule xii, clause 7),” 2025. Receipt and referral of measures, sponsorship and cosponsorship, and the constitutional authority statement requirement at clause 7(c)(1). <https://www.congress.gov/bill/119th-congress/house-resolution/5/text>.
- [42] Office of the Clerk, U.S. House, “ehopper quick guide (electronic introduction of measures),” 2026. The House primary electronic introduction mechanism; submissions accepted near convening and adjournment; generates and submits the constitutional authority statement. <https://www.house.gov/>.

- [43] Office of the Federal Register, “Quality management system regulation, 21 cfr part 820,” 2026. The codified device quality requirement, as amended by the Quality Management System Regulation effective Feb. 2, 2026. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-820>.
- [44] Office of the Parliamentarian, U.S. House, “How our laws are made,” 2026. The reference on the legislative process: introduction, referral, committee, and the bill versus resolution distinction. <https://www.congress.gov/help/learn-about-the-legislative-process/how-our-laws-are-made>.
- [45] Office of the Legislative Counsel, U.S. House, “Understanding the legislative process; house legislative counsel drafting guide,” 2026. Nonpartisan, confidential drafting services; positive-law status affects whether a bill amends the U.S. Code provision or the underlying statute. <https://legcounsel.house.gov/>.
- [46] Anthropic, “Claude code opus 4.8 (1m context) max,” 2026. The coding agent that authored the v2.0.0 to v2.3.0 pipeline autonomously across sequential commits; named for development transparency. <https://claude.ai/code>.
- [47] ASME, “Asme vvuq 70: Verification, validation, and uncertainty quantification for machine learning (in development),” 2026. The machine-learning credibility standard in development; cited as direction only. There is no ASME V&V 90. <https://www.asme.org/conferences-events/events/vvuq-70-verification-validation-and-uncertainty-quantification-of-machine-learning>.
- [48] International Organization for Standardization, “Iso 13482:2014: Robots and robotic devices - safety requirements for personal care robots,” 2014. Stability, collision avoidance, and fail-safe requirements for a close-contact humanoid. <https://www.iso.org/standard/53820.html>.
- [49] International Electrotechnical Commission, “Iec 80601-2-77:2019: Requirements for the basic safety and essential performance of robotically assisted surgical equipment,” 2019. The dedicated surgical-robot safety standard defining the essential performance the gate protects. <https://www.iso.org/standard/68473.html>.
- [50] IEEE, “Ieee std 7009-2024: Standard for fail-safe design of autonomous and semi-autonomous systems,” 2024. DOI: 10.1109/IEEESTD.2024.10582898. Fail-safe behavior and hand-back-to-human escalation underpinning the ESCALATE rule. <https://doi.org/10.1109/IEEESTD.2024.10582898>.
- [51] Underwriters Laboratories, “Ansi/ul 4600: Standard for safety for the evaluation of autonomous products,” 2023. The claim-based safety case (claims, argument, evidence) and safe-state default for the autonomy gates. <https://users.ece.cmu.edu/~koopman/ul4600/index.html>.
- [52] Congressional Research Service, “Introducing a house bill or resolution (crs report r44001),” 2025. The mechanics of introduction: the bill is not numbered until introduced, the Clerk assigns the number, and only a Member may introduce a House bill. <https://www.everycrsreport.com/reports/R44001.html>.
- [53] Office of the Law Revision Counsel, “United states code, title 21 - food and drugs (uslm xml),” 2026. Authoritative source of the reproduced existing-law text, current through Public Law 119-93. <https://uscode.house.gov>.

- [54] Congressional Research Service, “Sponsorship and cosponsorship of house bills (crs report rs22477),” 2024. One sponsor and an unlimited number of cosponsors; the basis for the sponsor and cosponsor packet deliverable. <https://crsreports.congress.gov/product/details?prodcode=RS22477>.
- [55] K. Kawchak, “Vvuq-05 update-bill next-steps: pre-submission steps for a u.s. house bill (2026),” 2026. Two multi-model outputs identifying the deliverables and process a draft House bill needs before introduction; the source for the deliverables appendix. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-05/update-bill/next-steps>.
- [56] K. Kawchak, “Verification before generation in physical ai oncology trials act of 2026 (h.r. 9510; fd&c act amendment, bill v3.0, the visual amendment),” June 2026. DOI: 10.5281/zenodo.20535429. Bill v3.0, this finished visual amendment; source at papers/VVUQ-05/full-bill. <https://doi.org/10.5281/zenodo.20535429>.
- [57] Executive Office of the President, “Executive order 14179, removing barriers to american leadership in artificial intelligence,” 2025. Research influence only (90 FR 8741); deregulatory posture cited in the findings narrative and Appendix A, never in operative text. <https://www.govinfo.gov/app/details/DCPD-202500170>.
- [58] Office of Management and Budget, “Omb memorandum m-25-21, accelerating federal use of ai through innovation, governance, and public trust,” 2025. Research influence only (Apr. 3, 2025); defines high-impact AI and pre-deployment testing for federal use. Cited only in Appendix A. <https://www.whitehouse.gov/omb/information-for-agencies/memoranda/>.
- [59] Executive Office of the President, “Executive order 14365, ensuring a national policy framework for artificial intelligence,” 2025. Research influence only (signed Dec. 11, 2025); directs review of conflicting State AI laws. Cited only in Appendix A. <https://public-inspection.federalregister.gov/2025-23092.pdf?1765815309>.
- [60] U.S. Department of Health and Human Services, “Hhs artificial intelligence strategy,” 2025. Research influence only (released Dec. 4, 2025); governance and research-reproducibility pillars cited in the findings narrative and Appendix A. <https://www.hhs.gov/programs/topic-sites/ai/index.html>.
- [61] United States Congress, “H.r. 238, healthy technology act of 2025,” 2025. Research influence only; 119th Congress bill on AI prescribing authority, referred to Energy and Commerce. Not operative authority. <https://www.congress.gov/bill/119th-congress/house-bill/238>.
- [62] United States Congress, “S. 1399, health tech investment act,” 2025. Research influence only; 119th Congress bill on a Medicare payment class for algorithm-based services, referred to Finance. Not operative. <https://www.congress.gov/bill/119th-congress/senate-bill/1399>.
- [63] United States Congress, “H.r. 6361, ban ai denials in medicare act,” 2025. Research influence only; 119th Congress bill barring AI-driven prior authorization in traditional Medicare. Not operative authority. <https://www.congress.gov/bill/119th-congress/house-bill/6361>.
- [64] United States Congress, “H.r. 5045, health ai act,” 2025. Research influence only; 119th Congress bill creating an HHS grant program on generative AI in health care, referred to Energy and Commerce. Not operative authority. <https://www.congress.gov/bill/119th-congress/house-bill/5045>.

- [65] United States Congress, “H.res. 694, expressing the sense of the house on the cms wiser artificial intelligence pilot,” 2025. Research influence only; a simple resolution urging CMS to halt AI in Medicare coverage decisions. Cited by number and subject; not operative authority. <https://www.congress.gov/bill/119th-congress/house-resolution/694>.
- [66] Office of the Law Revision Counsel, “Download the united states code,” 2026. Source portal for the Title 21 USLM XML reproduced in the comparative print (urldate 2026-05-31). <https://uscode.house.gov/download/download.shtml>.
- [67] United States Congress, “Food and drug omnibus reform act of 2022 (fdora), pub. l. 117-328, div. ff, title iii,” 2022. Added FD&C Act § 515C (21 U.S.C. § 360e-4), the PCP authority. <https://www.congress.gov/bill/117th-congress/house-bill/2617>.
- [68] United States Congress, “21st century cures act, pub. l. 114-255,” 2016. Source of the software and clinical decision support exclusion (21 U.S.C. § 360j(o)) and the real-world-evidence program (21 U.S.C. § 355g). <https://www.congress.gov/bill/114th-congress/house-bill/34>.
- [69] U.S. Food and Drug Administration, “Clinical decision support software (current guidance),” 2026. The controlling clinical decision support guidance; software that drives or executes a procedure cannot claim the 21 U.S.C. § 360j(o) exclusion. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software>.
- [70] Office of the Federal Register, “Premarket approval of medical devices, 21 cfr part 814,” 2026. The premarket-approval regulation under 21 U.S.C. § 360e implementing the application-contents. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-814>.
- [71] Office of the Federal Register, “Medical device classification procedures, 21 cfr part 860,” 2026. The classification regulation under 21 U.S.C. § 360c, including De Novo classification. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-860>.
- [72] Office for Civil Rights, HHS, “Nondiscrimination in health programs and activities (section 1557 final rule), 89 fr 37522,” 2024. Final rule (effective July 5, 2024) that promulgated 45 CFR § 92.210; the statutory authority is 42 U.S.C. § 18116. <https://www.govinfo.gov/ap/p/details/FR-2024-05-06/2024-08711>.
- [73] United States Congress, “Section 1557 of the patient protection and affordable care act, 42 u.s.c. § 18116,” 2010. The statutory nondiscrimination authority behind the 45 CFR § 92.210 rule. <https://www.law.cornell.edu/uscode/text/42/18116>.
- [74] United States Congress, “Medicare; exclusions from coverage (reasonable and necessary), 42 u.s.c. § 1395y(a)(1)(a),” 2026. A device cleared under this Act must still be reasonable and necessary for coverage; the verification record supports that determination. <https://www.law.cornell.edu/uscode/text/42/1395y>.
- [75] Assistant Secretary for Technology Policy, ONC, “Health data, technology, and interoperability: Certification program updates, algorithm transparency, and information sharing (hti-1), 89 fr 1192,” 2024. The decision-support intervention transparency template (45 CFR § 170.315(b)(11)); cited as algorithm-transparency context, not as operative authority. <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-C/section-170.315>.
- [76] United States Congress, “Section 504 of the rehabilitation act of 1973, 29 u.s.c. § 794,” 1973. Disability nondiscrimination basis (Title 29) complementing the accessibility duty. <https://www.law.cornell.edu/uscode/text/29/794>.

- [77] International Electrotechnical Commission, “Iec 62304:2006/amd 1:2015: Medical device software - software life cycle processes,” 2015. Software safety classification (class A, B, C); its sole normative reference is ISO 14971. <https://webstore.iec.ch/publication/22794>.
- [78] International Organization for Standardization, “Iso 10218-1 and -2: Robotics - safety requirements for industrial robots,” 2025. Speed-and-separation monitoring and safety-rated stops; the 2025 revision integrates the collaborative-robot limits of ISO/TS 15066. <https://www.iso.org/standard/73933.html>.
- [79] International Organization for Standardization, “Iso 9283:1998: Manipulating industrial robots - performance criteria and related test methods,” 1998. Pose accuracy and repeatability for the servo, grasp, and handover gates. <https://www.iso.org/standard/22244.html>.
- [80] National Institute of Standards and Technology, “Nist ai risk management framework 1.0 (nist ai 100-1),” 2023. DOI: 10.6028/NIST.AI.100-1. Risk-management framing (map, measure, manage, govern) as direction, not operative authority. <https://doi.org/10.6028/NIST.AI.100-1>.
- [81] Office of the Federal Register, “Institutional review boards, 21 cfr part 56,” 2026. The IRB review framework the bill preserves unchanged. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-56>.
- [82] K. Kawchak, “Vvuq physical ai oncology trial bill (h.r. 9510, draft 1.0),” May 2026. DOI: 10.5281/zenodo.20454870. The prior bill this amendment recasts; source at papers/VVUQ-03/final-paper. <https://doi.org/10.5281/zenodo.20454870>.
- [83] K. Kawchak, “cancer-automated: Automated physical ai oncology trial daily deliverables,” 2026. DOI: 10.5281/zenodo.20372501. The repository holding the VVUQ-01 through VVUQ-04 records and this full-bill. <https://github.com/kevinkawchak/cancer-automated>.
- [84] K. Kawchak, “physical-ai-oncology-trials: National platform,” 2026. The sibling repository with the USL and PSL standards and the trial simulations; read only as a source, not edited. <https://github.com/kevinkawchak/physical-ai-oncology-trials>.
- [85] K. Kawchak, “Vvuq-04 instruct-bill: U.s. medical ai bill and law summaries (through may 31, 2026),” 2026. The legal crosswalk and bill-style instruction set (v2.0.0) whose ten markdown files and five .bib files ground this bill. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/instruct-bill>.
- [86] K. Kawchak, “Vvuq-04 template-bill: Title 21 device-provisions excerpt (latex),” 2026. The faithful Title 21 reproduction (v2.1.0) whose eleven section files form the comparative-print base. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/template-bill>.
- [87] K. Kawchak, “Vvuq-04 draft-bill: Fd&c act amendment scaffold with drafting instructions (latex),” 2026. The amendment scaffold (v2.2.0) whose bracketed drafting instructions this full-bill executes. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/draft-bill>.
- [88] K. Kawchak, “Vvuq-03 final-paper: Vvuq physical ai oncology trial bill source,” 2026. The prior-bill LaTeX source whose operative text, definitions, and crosswalk this amendment converts into FD&C Act amendatory form. <https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-03/final-paper>.

- [89] New York State Legislature, “A9149, artificial intelligence in utilization review,” 2024. Would require submission of utilization-review algorithms for bias certification; the algorithm-certification model. Cited as a finding of existing practice. <https://www.nysenate.gov/legislation/bills/2023/A9149>.
- [90] K. Kawchak, “Verification before generation in physical ai oncology trials act of 2026 (h.r. 9510; fd&c act amendment, bill v2.0),” June 2026. DOI: 10.5281/zenodo.20485580. Bill v2.0, the finished FD&C Act amendment this visual bill reorganizes; source at papers/VVUQ-04/final-bill. <https://doi.org/10.5281/zenodo.20485580>.
- [91] K. Kawchak, “Two stage vvuv oncology clinical trial verification automation priority over existing generated code (vvuv-01),” 2026. DOI: 10.5281/zenodo.20372501. The method paper and pipeline; 51 of 51 tests, a 1 ACCEPT, 5 BLOCK, 1 ESCALATE gate surface, and the PDAC pilot reference. <https://doi.org/10.5281/zenodo.20372501>.
- [92] K. Kawchak, “Mobile pancreatic cancer unitree h2 surgical humanoid with priority vvuv (vvuv-02),” 2026. DOI: 10.5281/zenodo.20421754. The ten-gate humanoid assurance suite; 172 of 172 tests and a 32 of 32 sweep at composite mean 93.56 from seed 20260525. <https://doi.org/10.5281/zenodo.20421754>.
- [93] Congressional Research Service, “Constitutional authority statements: A quick guide (crs report if12335),” 2023. Required by House Rule XII, clause 7(c); for an FD&C Act amendment the cited power is the Commerce Clause (Art. I, sec. 8, cl. 3). <https://crsreports.congress.gov/product/details?prodcode=IF12335>.
- [94] National Institute of Standards and Technology, “Nist ai 600-1: Artificial intelligence risk management framework: Generative ai profile,” 2024. DOI: 10.6028/NIST.AI.600-1. Generative-AI risk and transparency framing; direction for the explainability standard, not operative authority. <https://doi.org/10.6028/NIST.AI.600-1>.
- [95] P. J. Phillips, C. A. Hahn, P. C. Fontana, A. N. Yates, K. Greene, D. A. Broniatowski, and M. A. Przybocki, “Four principles of explainable artificial intelligence (nistir 8312),” 2021. DOI: 10.6028/NIST.IR.8312. The explanation, meaningful, accuracy, and knowledge-limits principles the four-work record is mapped to in Appendix C. <https://doi.org/10.6028/NIST.IR.8312>.